Motivic Homotopy Theory (Draft)

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Preface

These notes grew out of my participation in Marc Hoyois's Oberseminar on motivic cohomology at the University of Regensburg (SoSe 2026), combined with my own study of the subject. The primary references are the work of Bachmann–Hoyois [BH21] on norms, and the recent paper of Bachmann–Elmanto–Morrow [BEM25].

The goal is to give a self-contained introduction to motivic homotopy theory.

Warning. *This document is a preliminary draft. It likely contains errors, incomplete arguments, and missing references. Corrections and suggestions are welcome.*

A Chinese version of part of this material is available on Banana Space.

Conventions

Throughout these notes, we work in the framework of ∞-categories. The word "category" means ∞-category unless otherwise specified; we write "ordinary category" or "1-category" for the classical notion.

All schemes are assumed to be quasi-compact and quasi-separated unless otherwise stated.

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Chapter 1.

Unstable Motivic Homotopy Theory

1.1. Introduction and Motivation

The starting point of motivic homotopy theory is a striking analogy between algebraic geometry and topology. In topology, for any paracompact Hausdorff space X, the projection $X \times I \to X$ (with $I = [0,1]$) induces a bijection

$$\pi_0 \mathsf{Vect}_r^{\mathrm{top}}(X) \xrightarrow{\sim} \pi_0 \mathsf{Vect}_r^{\mathrm{top}}(X \times I).$$

Vector bundles, and many other invariants, are insensitive to the cylinder direction. The classical Quillen–Suslin theorem [Qui76, Sus76] provides the algebraic counterpart: every finitely generated projective module over $k[x_1, \dots, x_n]$ is free, equivalently, the pullback

$$\pi_0 \mathsf{Vect}_r(\mathrm{Spec}(k)) \xrightarrow{\sim} \pi_0 \mathsf{Vect}_r(\mathbb{A}^n_k)$$

is a bijection. Under this analogy, the affine line $\mathbb{A}^1$ plays the role of the interval I: both serve as "cylinder objects" in their respective categories, suggesting a homotopy theory for algebraic geometry in which the projection $\mathbb{A}^1 \times X \to X$ is declared a homotopy equivalence.

Supporting this intuition, many cohomology theories on smooth schemes are $\mathbb{A}^1$-invariant. For a smooth scheme X over a field k:

- **Chow groups:** $\mathrm{CH}^*(X) \xrightarrow{\sim} \mathrm{CH}^*(X \times \mathbb{A}^1)$;
- **Grothendieck group:** $\mathrm{K}_0(X) \xrightarrow{\sim} \mathrm{K}_0(X \times \mathbb{A}^1)$;
- **Étale cohomology** (for ℓ invertible in k): $\mathrm{H}^*_{\text{ét}}(X, \mu_\ell) \xrightarrow{\sim} \mathrm{H}^*_{\text{ét}}(X \times \mathbb{A}^1, \mu_\ell)$.

***Remark* 1.1.1** ($\mathbb{A}^1$-invariance fails on non-affine schemes)**.** The functor $\pi_0 \mathsf{Vect}_r \colon \mathsf{Sm}_k^{\mathrm{op}} \to \mathsf{Set}$ is *not* $\mathbb{A}^1$-invariant in general. Take the rank-2 bundle E on $\mathbb{P}^1 \times \mathbb{A}^1$ obtained by gluing the trivial bundle on the standard charts $U_0 = \mathrm{Spec}(k[x]) \times \mathbb{A}^1$ and $U_1 = \mathrm{Spec}(k[x^{-1}]) \times \mathbb{A}^1$ via the transition matrix

$$g_{01}(x,t) = \begin{pmatrix} x & t \\ 0 & x^{-1} \end{pmatrix} \in \mathrm{GL}_2\big(k[x, x^{-1}, t]\big).$$

At $t = 0$, the matrix is diagonal, giving $E|_{\mathbb{P}^1 \times \{0\}} \simeq \mathcal{O}(1) \oplus \mathcal{O}(-1)$. At $t = 1$, the matrix defines a non-trivial extension

$$0 \longrightarrow \mathcal{O}(-1) \longrightarrow E|_{\mathbb{P}^1 \times \{1\}} \longrightarrow \mathcal{O}(1) \longrightarrow 0$$

whose class in $\mathrm{Ext}^1(\mathcal{O}(1), \mathcal{O}(-1)) = H^1(\mathbb{P}^1, \mathcal{O}(-2)) \cong k$ is non-zero; by Grothendieck's splitting theorem, this extension is isomorphic as a vector bundle to $\mathcal{O} \oplus \mathcal{O}$. Since $E|_{\mathbb{P}^1 \times \{0\}} \not\simeq E|_{\mathbb{P}^1 \times \{1\}}$ on $\mathbb{P}^1$, the bundle E cannot lie in the image of $\pi_0 \mathsf{Vect}_2(\mathbb{P}^1) \to \pi_0 \mathsf{Vect}_2(\mathbb{P}^1 \times \mathbb{A}^1)$.

Nevertheless—and this is the deep theorem of Morel [Mor12], simplified by Asok–Hoyois–Wendt [AHW17]—if we forcibly $\mathbb{A}^1$-localize Vect_r, the resulting functor still computes the correct values on *smooth affine* schemes. This affine representability theorem is one of the cornerstone applications of motivic homotopy theory.

1.1.1. Overview of the construction

Fix a quasi-compact quasi-separated scheme S, and let Sm_S denote the category of smooth S-schemes of finite presentation. Our goal is to construct the **motivic homotopy category** $\mathsf{H}(S)$: an enlargement of Sm_S in which the projections $\mathbb{A}^1 \times_S X \to X$ become equivalences. The construction proceeds in three universal steps, each with a clear geometric meaning:

(i) **Sifted cocompletion.** Embed Sm_S into $\mathcal{P}_\Sigma(\mathsf{Sm}_S)$, the category of presheaves of anima that convert finite coproducts to finite products.

(ii) **Nisnevich descent.** Force a chosen class of étale-like covers (the *Nisnevich covers*) to become equivalences, yielding $\mathsf{Shv}_{\mathrm{Nis}}(\mathsf{Sm}_S)$.

(iii) **$\mathbb{A}^1$-localization.** Force the projections $\mathbb{A}^1 \times X \to X$ to become equivalences, yielding the motivic homotopy category $\mathsf{H}(S)$.

***Remark* 1.1.2** (Why $\mathcal{P}_\Sigma$ rather than PShv?)**.** One could equally well start from the full presheaf category $\mathsf{PShv}(\mathsf{Sm}_S)$. The advantage of $\mathcal{P}_\Sigma(\mathsf{Sm}_S)$ is threefold:

(i) Since Sm_S is extensive, $\mathcal{P}_\Sigma(\mathsf{Sm}_S) \simeq \mathsf{Shv}_\sqcup(\mathsf{Sm}_S)$ [BH21, Lemma 2.4], which is already a topos for the coproduct topology.

(ii) The Yoneda embedding $\mathsf{Sm}_S \hookrightarrow \mathcal{P}_\Sigma(\mathsf{Sm}_S)$ preserves finite coproducts, and every object is a sifted colimit of representables.

(iii) The universal property of $\mathcal{P}_\Sigma$ as the free sifted cocompletion is essential to the construction of norm functors in §1.7; see [BH21, §3].

Since finite coproduct decompositions are themselves Nisnevich covers, we have natural inclusions

$$\mathsf{Shv}_{\mathrm{Nis}}(\mathsf{Sm}_S) \subset \mathcal{P}_\Sigma(\mathsf{Sm}_S) \subset \mathsf{PShv}(\mathsf{Sm}_S),$$

and we write $\mathrm{L}_{\mathrm{Nis}} : \mathcal{P}_\Sigma(\mathsf{Sm}_S) \to \mathsf{Shv}_{\mathrm{Nis}}(\mathsf{Sm}_S)$ for the Nisnevich sheafification functor.

1.2. The Nisnevich Topology

The Nisnevich topology, also called the completely decomposed topology, sits strictly between Zariski and étale. In motivic homotopy theory, this choice is forced by two complementary optimality conditions: it is the coarsest topology for which the purity theorem holds, and the finest for which algebraic K-theory is representable (Remark 1.2.1).

1.2.1. Three topologies

For S qcqs, there are three commonly used Grothendieck topologies on Sm_S:

$$\mathrm{Zar} \subset \mathrm{Nis} \subset \text{ét}.$$

- **Zariski:** covers are jointly surjective open immersions.
- **Étale:** covers are jointly surjective étale morphisms.
- **Nisnevich:** covers are jointly surjective étale morphisms with the additional requirement that no residue field extensions occur.

The functor-of-points perspective makes the Nisnevich condition transparent. For a smooth k-scheme X, an étale cover $\{U_i \to X\}$ need not satisfy $\coprod_i U_i(k) \twoheadrightarrow X(k)$: the preimage of a k-rational point might only exist over a finite separable extension L/k. A Nisnevich cover excludes precisely this phenomenon—every k-rational point lifts without field extension.

***Remark* 1.2.1** (Why Nisnevich?)**.** The Nisnevich topology is characterized by two complementary optimality conditions:

- **Coarsest topology for which purity holds.** The purity theorem asserts that for a closed immersion $Z \hookrightarrow X$ in Sm_S with normal bundle ν_Z, there is an $\mathbb{A}^1$-local equivalence $X/(X \setminus Z) \simeq \mathrm{Th}(\nu_Z)$. The Zariski topology is too coarse for this.

- **Finest topology for which K-theory is representable.** Non-connective K-theory $\mathrm{K}\colon \mathsf{Sm}_S^{\mathrm{op}} \to \mathsf{Sp}$ satisfies Nisnevich descent but not étale descent.

1.2.2. Nisnevich covers

Definition 1.2.2 (Nisnevich cover). A family of étale morphisms $(f_i\colon U_i \to X)_{i\in I}$ in Sm_S is a **Nisnevich cover** if for every $x \in X$, there exists $i \in I$ and $u \in U_i$ with $f_i(u) = x$ and $\kappa(u) \xrightarrow{\sim} \kappa(x)$.

***Remark* 1.2.3.** The term "completely decomposed" originates in arithmetic: a prime ideal $\mathfrak{p}$ of $\mathcal{O}_K$ is "completely decomposed" in an étale cover precisely when its residue field does not extend.

Theorem 1.2.4 (Equivalent characterizations). *Let X be qcqs and $(f_i\colon U_i \to X)_{i\in I}$ a family of étale morphisms. The following are equivalent:*

(i) ***(Rational points)*** *For every $x \in X$, there exists i and $u \in U_i$ with $f_i(u) = x$ and $\kappa(u) \xrightarrow{\sim} \kappa(x)$.*

(ii) ***(Finite stratification)*** *There is a chain of finitely presented closed subschemes*

$$\varnothing = Z_0 \subset Z_1 \subset \cdots \subset Z_n = X$$

such that some f_i admits a section over each stratum $Y_j = Z_j \setminus Z_{j-1}$.

(iii) ***(Refinement by Nisnevich squares)*** *There is a stratification as above and a finitely presented étale family $(g_j\colon V_j \to X)_{j=1}^n$ refining (f_i), with each g_j restricting to an isomorphism over Y_j.*

Proof. The implications (iii) ⇒ (ii) ⇒ (i) are immediate. We prove (i) ⇒ (iii).

Replacing each U_i by an affine open cover, we may assume all U_i are quasi-compact and separated, hence all f_i are of finite presentation. Consider the collection of closed subschemes $Y \subseteq X$ for which (iii) fails; we show this collection is empty by Zorn's lemma.

Let $(Y_\alpha)_{\alpha\in A}$ be a descending chain and $Y = \bigcap_\alpha Y_\alpha$. If (iii) holds for Y, then by finite presentation it holds for some Y_α (limits of finitely presented data along cofiltered systems). Hence if (iii) fails, there exists a minimal closed $Y \subseteq X$ for which it fails. Replacing X by Y, we may assume (iii) fails for X but holds for all proper closed subschemes.

Clearly $X \neq \varnothing$. Take a generic point $x \in X$. By (i), for some i there is a morphism $\mathrm{Spec}(\kappa(x)) \to U_i$ lifting $\mathrm{Spec}(\kappa(x)) \to X$. Since $\mathcal{O}_{X,x}^{\mathrm{red}} = \kappa(x)$, the nilpotent lifting property of étale morphisms yields a lift $\mathrm{Spec}(\mathcal{O}_{X,x}) \to U_i$. Passing to a limit, we obtain a quasi-compact open neighborhood $V \subseteq X$ of x and a section $s\colon V \to U_i$.

Take a finitely presented closed $Z \subset X$ with $V = X \setminus Z$. By minimality, (iii) holds for Z: there is a stratification $\varnothing = Z_0 \subset \cdots \subset Z_n = Z$ and a refining family $(h_j\colon W_j \to Z)$ with the required properties. For each j, lift W_j to a finitely presented open $V_j \subseteq U_{i_j}$ with $g_j\colon V_j \to X$ étale and an isomorphism over Y_j. Extending the chain to $\varnothing = Z_0 \subset \cdots \subset Z_n \subset Z_{n+1} = X$ with $V_{n+1} = V$ and $g_{n+1} = f_i \circ s$ realizes (iii) for X, contradicting minimality. □

1.2.3. Nisnevich squares

The key feature of the Nisnevich topology in practice is that it is generated by a particularly simple class of covers.

Definition 1.2.5 (Nisnevich square). A pullback square in Sm_S

$$\begin{tikzcd} U \times_X V \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & V \arrow[d, "p"] \\ U \arrow[r, hook, "j"'] & X \end{tikzcd}$$

is a **Nisnevich square** if

- j is an open immersion;
- p is étale;

- p induces an isomorphism $p^{-1}(X \setminus U)_{\mathrm{red}} \xrightarrow{\sim} (X \setminus U)_{\mathrm{red}}$.

From the functor-of-points perspective, a Nisnevich square decomposes X's R-points into two classes: those landing in the open U, handled by j; and those in the closed complement $X \setminus U$, covered by p without any residue field extension.

Lemma 1.2.6. *The family $\{j\colon U \hookrightarrow X,\ p\colon V \to X\}$ is a Nisnevich cover of X.*

Proof. For $x \in U$, j has x in its image with trivial residue extension. For $x \in X \setminus U$, the assumption gives $v \in p^{-1}(x)$ with $\kappa(v) \simeq \kappa(x)$. □

The deeper result is that Nisnevich squares *generate* the Nisnevich topology:

Proposition 1.2.7 (Brown–Gersten)**.** *A presheaf $\mathcal{F}$ on Sm_S is a Nisnevich sheaf if and only if $\mathcal{F}(\varnothing)$ is terminal and $\mathcal{F}$ sends every Nisnevich square to a pullback square:*

$$\mathcal{F}(X) \xrightarrow{\sim} \mathcal{F}(V) \times_{\mathcal{F}(U \times_X V)} \mathcal{F}(U).$$

Proof. ($\Rightarrow$) By Lemma 1.2.6, $\{U \hookrightarrow X, V \to X\}$ is a Nisnevich cover, and the sheaf condition directly yields the pullback condition. The empty cover gives $\mathcal{F}(\varnothing) = *$.

($\Leftarrow$) Assume $\mathcal{F}$ sends Nisnevich squares to pullbacks. We show $\mathcal{F}$ satisfies the sheaf condition for any Nisnevich cover of X. By Theorem 1.2.4(iii), there is a refining family $(g_j\colon V_j \to X)_{j=1}^n$ with stratification $\varnothing = Z_0 \subset \cdots \subset Z_n = X$ and each g_j an isomorphism over Y_j. Induct on n.

For $n = 0$, $X = \varnothing$ and $\mathcal{F}(\varnothing) = *$ gives the result. For $n \geq 1$, set $U = X \setminus Z_1$, so $j\colon U \hookrightarrow X$ is open with $X \setminus U = Z_1$. Since g_1 is an isomorphism over Z_1, this yields a Nisnevich square, giving

$$\mathcal{F}(X) \simeq \mathcal{F}(V_1) \times_{\mathcal{F}(U \times_X V_1)} \mathcal{F}(U).$$

The induced stratification on U has length $n-1$, so by induction $\mathcal{F}$ is a sheaf on U (and on $U \times_X V_1$). Transitivity of pullbacks finishes the proof. □

***Warning* 1.2.8** (Non-subcanonicality)**.** The Nisnevich topology is *not* subcanonical: representable presheaves $y(X) = \mathrm{Hom}_{\mathsf{Sm}_S}(-, X)$ are in general not Nisnevich sheaves. Concretely, Nisnevich descent for h_X would require X-points to glue along Nisnevich covers, but this already fails for henselizations: the henselian local ring $\mathcal{O}^h_{Y,y}$ is the colimit of étale neighborhoods of y, yet $X(\mathcal{O}^h_{Y,y})$ generally does *not* agree with $\operatorname{colim} X(U)$ over those neighborhoods unless X is sufficiently rigid.

Consequently, throughout the construction of $\mathsf{H}(S)$ we must Nisnevich-sheafify representables before working with them. The same warning applies, a fortiori, to the cdh topology of §1.3.

***Remark* 1.2.9** (Finitarity)**.** The Nisnevich and cdh topologies are both *finitary*: every covering family may be refined to a finite subcover. This is essential for several compatibility results we will use:

- Nisnevich and cdh sheafification commute with filtered colimits.
- The topoi $\mathsf{Shv}_{\mathrm{Nis}}(\mathsf{Sm}_S)$ and $\mathsf{Shv}_{\mathrm{cdh}}(\mathsf{Sch}_S^{\mathrm{fp}})$ are compactly generated, with compact generators given by the (sheafifications of) representables.
- Hypercompletion behaves well, and on finite-dimensional schemes coincides with the topos itself by Theorems 1.2.15 and 1.3.6.

1.2.4. Examples

Example 1.2.10 (Zariski covers are Nisnevich)**.** Open immersions induce identity maps on residue fields, so Zariski covers are automatically Nisnevich. The standard cover $\{\mathbb{A}^1, \mathbb{A}^1\} \to \mathbb{P}^1$ corresponds to the Nisnevich square

$$\begin{tikzcd} \mathbb{G}_m \arrow[r] \arrow[d] & \mathbb{A}^1 \arrow[d] \\ \mathbb{A}^1 \arrow[r] & \mathbb{P}^1. \end{tikzcd}$$

Example 1.2.11 (The m-th power map). Let k be a field with $\mathrm{char}(k) \nmid m$ and $a \in k^\times$. Consider

$$\begin{aligned} f\colon (\mathbb{A}^1 \setminus \{a\}) \sqcup (\mathbb{A}^1 \setminus \{0\}) &\longrightarrow \mathbb{A}^1, \\ \text{first component:} \quad x &\longmapsto x, \\ \text{second component:} \quad y &\longmapsto y^m. \end{aligned}$$

The first component is an open immersion; the second is étale. This f is a Nisnevich cover if and only if a has an m-th root in k.

Example 1.2.12 (Algebraic K-theory is a Nisnevich sheaf). By Brown–Gersten it suffices to check Nisnevich excision (the empty case is trivial). For a Nisnevich square

$$\begin{tikzcd} W \arrow[r] \arrow[d] & V \arrow[d] \\ U \arrow[r, "j"'] & Y, \end{tikzcd}$$

perfect complexes satisfy flat descent, so we obtain a pullback square of stable categories

$$\begin{tikzcd} \mathsf{Perf}(W) & \mathsf{Perf}(V) \arrow[l] \\ \mathsf{Perf}(U) \arrow[u] & \mathsf{Perf}(Y). \arrow[l] \arrow[u] \end{tikzcd}$$

Setting $Z = Y \setminus U = V \setminus W$ and taking horizontal fibers gives a diagram of Karoubi sequences (using full faithfulness of $j_*\colon \mathsf{D}(U) \to \mathsf{D}(Y)$). Since K-theory sends Karoubi sequences to fiber sequences of spectra, the K-theory square is also a pullback.

1.2.5. Milnor excision

Restricting to affine schemes, Nisnevich squares take a particularly clean ring-theoretic form.

Definition 1.2.13 (Milnor square). A **Milnor square** for a commutative ring R is a pullback

$$\begin{tikzcd} R \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & R/I \arrow[d] \\ R_f \arrow[r] & R_f/IR_f \end{tikzcd}$$

where $f \in R$, $I \subset R$ is an ideal, and $R/I \xrightarrow{\sim} R_f/IR_f$ (equivalently, f is a unit in R/I).

A Milnor square is precisely a Nisnevich square for affine schemes: $U = \mathrm{Spec}(R_f) \hookrightarrow \mathrm{Spec}(R) = X$ is the principal open defined by f, and $V = \mathrm{Spec}(R/I) \to X$ is a closed immersion. The condition $R/I \simeq R_f/IR_f$ is exactly $p^{-1}(X \setminus U) \simeq X \setminus U$.

Proposition 1.2.14 (Milnor excision). *For every Nisnevich sheaf $\mathcal{F}$ and every Milnor square,*

$$\mathcal{F}(\mathrm{Spec}(R)) \xrightarrow{\sim} \mathcal{F}(\mathrm{Spec}(R_f)) \times_{\mathcal{F}(\mathrm{Spec}(R_f/IR_f))} \mathcal{F}(\mathrm{Spec}(R/I)).$$

1.2.6. Cohomological dimension of the Nisnevich topos

A foundational structural result about the Nisnevich topology is a sharp bound on the homotopy dimension of its associated topos. Recall that an ∞-topos $\mathcal{X}$ has **homotopy dimension** $\leq d$ if every $(d-1)$-connected sheaf on $\mathcal{X}$ admits a global section. Equivalently, sheaves of n-truncated anima "can be controlled" once $n \geq d$.

Theorem 1.2.15 (Clausen–Mathew). *For any qcqs scheme X of finite Krull dimension,*

$$\dim \mathsf{Shv}_{\mathrm{Nis}}(\mathsf{Et}_X) \;\leq\; \mathrm{Kr.dim}(X),$$

where the left-hand side denotes the homotopy dimension of the Nisnevich topos on the small étale site of X, and $\mathrm{Kr.dim}(X)$ is the Krull dimension.

***Remark* 1.2.16** (Geometric characterization of Nisnevich-local schemes)**.** A qcqs scheme X has no non-trivial Nisnevich coverings if and only if X is henselian local. This explains why Nisnevich stalks are computed at henselian local rings: the strict henselization $\mathcal{O}^h_{X,x}$ at a point $x \in X$ realizes the Nisnevich-local model of an étale neighborhood of x. Combined with Theorem 1.2.15, this gives a tight geometric grip on $\mathsf{Shv}_{\mathrm{Nis}}$: sheaves are determined by their values on henselian local rings, and the Postnikov tower converges in finite Krull dimension.

This is a far-reaching generalization of the classical fact, due to Grothendieck and Kato–Saito, that Nisnevich cohomology vanishes above the Krull dimension. The ∞-categorical reformulation given by Clausen–Mathew is more powerful: it controls not only cohomology groups but also the truncation behavior of arbitrary sheaves of anima. In particular, on a finite-dimensional scheme, every Nisnevich sheaf of spectra is hypercomplete, and every Postnikov tower converges.

***Remark* 1.2.17** (Comparison with étale and Zariski)**.** The bound in Theorem 1.2.15 is sharp and reflects another structural advantage of the Nisnevich topology over its neighbors:

- **Zariski:** $\dim \mathsf{Shv}_{\mathrm{Zar}}(X) \leq \mathrm{Kr.dim}(X)$ as well, but the Zariski topology is too coarse for many cohomology theories (e.g., it does not see Galois descent).
- **Étale:** no such finite bound holds in general; the étale topos can have infinite cohomological dimension (e.g., for $\mathrm{Spec}(k)$ with k a field of positive characteristic and large absolute Galois group).

The Nisnevich topology thus combines the dimensional finiteness of Zariski with enough geometric refinement to support algebraic K-theory and motivic constructions.

1.3. The cdh Topology

In both Nisnevich and étale topologies, covering morphisms are étale, hence local homeomorphisms in the algebraic sense: they preserve dimension and tangent spaces. Consequently, *étale covers cannot resolve singularities*.

Recall the nodal curve $N = \mathrm{Spec}(k[x,y]/(y^2 - x^3 - x^2))$. The normalization $\varphi\colon \tilde{N} \to N$ is an isomorphism away from the origin but collapses two points to one at the origin—and is certainly not étale. To unify the study of singular and smooth schemes, we need a finer topology that treats blowups (and more generally, proper birational morphisms) as covers. This is the **cdh topology** (complete decomposition with h-descent).

1.3.1. Abstract blowup squares

Definition 1.3.1 (Abstract blowup square)**.** A commutative diagram in Sch_S

$$\begin{tikzcd} E \arrow[r] \arrow[d, "q"'] \arrow[dr, phantom, "\lrcorner", very near start] & \tilde{X} \arrow[d, "p"] \\ Z \arrow[r, hook, "i"'] & X \end{tikzcd}$$

is an **abstract blowup square** if

- i is a closed immersion;
- p is proper;
- p induces an isomorphism $p^{-1}(X \setminus Z)_{\mathrm{red}} \xrightarrow{\sim} (X \setminus Z)_{\mathrm{red}}$;
- $E = \tilde{X} \times_X Z$ (as a reduced scheme).

***Remark* 1.3.2** (Comparison with Nisnevich squares)**.** Abstract blowup squares are structurally parallel to Nisnevich squares:

	Nisnevich square	**Abstract blowup square**
Decomposition of X	$U \overset{\text{open}}{\hookrightarrow} X$	$Z \overset{\text{closed}}{\hookrightarrow} X$
Covering morphism	$V \to X$ étale	$\tilde{X} \to X$ proper
Isomorphism away from	$X \setminus U$	$X \setminus Z$
Typical source	Étale neighborhoods	Blowups, normalizations

Both decompose X into a known part and a part needing coverage, then handle the latter via the appropriate class of morphisms.

Example 1.3.3 (cdh cover of the nodal curve). For the nodal curve N with $\mathrm{char}(k) \neq 2$:

$$\begin{tikzcd} E \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \tilde{N} \simeq \mathbb{A}^1 \arrow[d, "p"] \\ Z = \{0\} \arrow[r, hook] & N, \end{tikzcd}$$

where p is the normalization (finite, hence proper) and $E = \mathrm{Spec}(k[c]/(c^2 - 1)) \simeq \mathrm{Spec}(k) \sqcup \mathrm{Spec}(k)$ encodes the two tangent directions at the node. For a cdh sheaf $\mathcal{F}$,

$$\mathcal{F}(N) \simeq \mathcal{F}(\mathbb{A}^1) \times_{\mathcal{F}(\{c=1\} \sqcup \{c=-1\})} \mathcal{F}(\{0\}).$$

Thus invariants of the singular curve N are completely determined by their values on the smooth model $\mathbb{A}^1$, the singular point $\{0\}$, and the gluing data over E.

1.3.2. Definition of the cdh topology

Definition 1.3.4 (cdh topology). For X qcqs, the **cdh topology** on $\mathsf{Sch}_X^{\mathrm{qcqs}}$ is the Grothendieck topology generated by

(i) Nisnevich squares (Definition 1.2.5), and

(ii) abstract blowup squares (Definition 1.3.1).

A presheaf $\mathcal{F}$ on $\mathsf{Sch}_X^{\mathrm{qcqs}}$ is a **cdh sheaf** if it sends both types of squares to pullback squares—equivalently, satisfies both Nisnevich excision and proper excision.

1.3.3. Cohomological dimension of the cdh topos

Just as the Nisnevich topos has homotopy dimension bounded by the Krull dimension (Theorem 1.2.15), the cdh topos admits a parallel bound, this time in terms of the valuation dimension. Recall:

Definition 1.3.5 (Valuation dimension). The **valuation dimension** $\mathrm{v.dim}(X)$ of a qcqs scheme X is defined by $\mathrm{v.dim}(X) \leq d$ if and only if for every integral affine subscheme $\mathrm{Spec}(A) \subseteq X$ and every chain

$$A \subseteq V_0 \subseteq V_1 \subseteq \cdots \subseteq V_n \subseteq \mathrm{Frac}(A)$$

of valuation subrings of the fraction field with strict inclusions, one has $n \leq d$.

The valuation dimension always satisfies $\mathrm{v.dim}(X) \geq \mathrm{Kr.dim}(X)$, with equality when X is Noetherian or, more generally, when X is a valuative Noetherian scheme. The two notions coincide on most schemes one encounters in practice.

Theorem 1.3.6 (Elmanto–Hoyois–Iwasa–Kelly). *For any qcqs scheme X,*

$$\dim \mathsf{Shv}_{\mathrm{cdh}}(\mathsf{Sch}_X^{\mathrm{fp}}) \;\leq\; \mathrm{v.dim}(X),$$

where the left-hand side is the homotopy dimension of the cdh topos on the category of finitely presented X-schemes.

***Remark* 1.3.7** (Why valuation dimension?). The cdh topology is sensitive to phenomena invisible to the Nisnevich topology—namely, those involving non-Noetherian behavior such as the structure of valuation rings of fraction fields. The valuation dimension captures exactly the right invariant. Two key consequences:

(i) **$\mathbb{A}^1$-stability:** $\mathrm{v.dim}(X \times \mathbb{A}^n) = \mathrm{v.dim}(X) + n$, mirroring the analogous Krull dimension formula but more robustly behaved on non-Noetherian schemes.

(ii) **Coincidence with Krull dimension:** $\mathrm{v.dim}(X) = \mathrm{Kr.dim}(X)$ when X is Noetherian, or more generally a valuation ring.

On Noetherian schemes, Theorem 1.3.6 thus specializes to a Krull-dimensional bound, paralleling the Clausen–Mathew result for the Nisnevich topos. The genuinely new content is the extension to non-Noetherian schemes via valuation theory.

***Remark* 1.3.8** (Significance for motivic cohomology). The bound in Theorem 1.3.6 is the key technical input for many recent advances in motivic cohomology over singular and non-Noetherian bases, including the construction of motivic cohomology of arbitrary qcqs schemes by Elmanto–Morrow [EM23], and the comparison with prismatic cohomology in Bachmann–Elmanto–Morrow [BEM25]. Without dimensional finiteness, Postnikov towers in the cdh topos would not converge, and the cdh-local replacement of motivic invariants would be intractable.

1.3.4. Extension from smooth to singular schemes

The deepest application of the cdh topology is the systematic extension of cohomology theories from Sm_S to $\mathsf{Sch}_S^{\mathrm{qcqs}}$.

Construction 1.3.9 (Left Kan extension followed by cdh sheafification). Let $\mathcal{F} : \mathsf{Sm}_S^{\mathrm{op}} \to \mathsf{Sp}$ be a presheaf of spectra. Extend $\mathcal{F}$ to all qcqs S-schemes in two steps:

(i) **Left Kan extension along $\mathsf{Sm}_S \hookrightarrow \mathsf{Sch}_S^{\mathrm{qcqs}}$:**

$$\begin{tikzcd} \mathsf{Sm}_S^{\mathrm{op}} \arrow[r, "\mathcal{F}"] \arrow[d, hook] & \mathsf{Sp} \\ (\mathsf{Sch}_S^{\mathrm{qcqs}})^{\mathrm{op}} \arrow[ur, dashed, "\mathrm{Lan}^{\mathrm{sm}}\mathcal{F}"'] & \end{tikzcd}$$

Explicitly, $(\mathrm{Lan}^{\mathrm{sm}}\mathcal{F})(Y) = \mathrm{colim}_{(U \to Y) \in ((\mathsf{Sm}_S)_{/Y})^{\mathrm{op}}} \mathcal{F}(U)$.

(ii) **cdh sheafification:**

$$\mathcal{F}_{\mathrm{cdh}} \coloneqq \mathrm{L}_{\mathrm{cdh}}\big(\mathrm{Lan}^{\mathrm{sm}}\mathcal{F}\big).$$

When $\mathcal{F}$ is already a Nisnevich sheaf on smooth schemes, $\mathcal{F}_{\mathrm{cdh}}$ agrees with $\mathcal{F}$ on smooth schemes but acquires correct values on singular schemes via cdh descent.

1.4. $\mathbb{A}^1$-Localization

Write $\mathsf{PShv}_{\mathbb{A}^1}(\mathsf{Sm}_S) \subset \mathsf{PShv}(\mathsf{Sm}_S)$ for the full subcategory of **$\mathbb{A}^1$-invariant** presheaves: those $\mathcal{F}$ with $\mathcal{F}(U) \xrightarrow{\sim} \mathcal{F}(U \times_S \mathbb{A}^1)$ for all $U \in \mathsf{Sm}_S$. The inclusion admits a left adjoint

$$\mathrm{L}_{\mathbb{A}^1} : \mathsf{PShv}(\mathsf{Sm}_S) \longrightarrow \mathsf{PShv}_{\mathbb{A}^1}(\mathsf{Sm}_S),$$

called the **$\mathbb{A}^1$-localization functor**.

Definition 1.4.1 ($\mathbb{A}^1$-local equivalence and $\mathbb{A}^1$-homotopy). Let $f : \mathcal{F} \to \mathcal{G}$ be a morphism in $\mathsf{PShv}(\mathsf{Sm}_S)$.

- f is an **$\mathbb{A}^1$-local equivalence** if $\mathrm{L}_{\mathbb{A}^1}(f)$ is an isomorphism.
- Two maps $f, g : \mathcal{F} \to \mathcal{G}$ are **$\mathbb{A}^1$-homotopic** if there exists $H : \mathcal{F} \times \mathbb{A}^1 \to \mathcal{G}$ with $H \circ i_0 \simeq f$ and $H \circ i_1 \simeq g$, where $i_s : * \to \mathbb{A}^1$ is the inclusion of $s \in \{0, 1\}$.

- f is an $\mathbb{A}^1$**-homotopy equivalence** if there exists g with fg and gf each $\mathbb{A}^1$-homotopic to the appropriate identity.

Proposition 1.4.2. *Every $\mathbb{A}^1$-homotopy equivalence is an $\mathbb{A}^1$-local equivalence. In particular, the projection $p\colon \mathcal{F} \times \mathbb{A}^1 \to \mathcal{F}$ is an $\mathbb{A}^1$-local equivalence for every $\mathcal{F}$.*

Proof. We first show $p\colon X \times \mathbb{A}^1 \to X$ is an $\mathbb{A}^1$-homotopy equivalence for every $X \in \mathsf{Sm}_S$ (and hence, by universal colimits in PShv, for every presheaf). The zero section $s\colon X \to X \times \mathbb{A}^1$ satisfies $p \circ s = \mathrm{id}_X$. The reverse composite $s \circ p$ is $\mathbb{A}^1$-homotopic to $\mathrm{id}_{X\times\mathbb{A}^1}$ via

$$(X \times \mathbb{A}^1) \times \mathbb{A}^1 \xrightarrow{\mathrm{id}_X \times \mu} X \times \mathbb{A}^1,$$

where $\mu\colon \mathbb{A}^1 \times \mathbb{A}^1 \to \mathbb{A}^1$ is the ring multiplication: at $t = 1$ this gives $\mathrm{id}_{X\times\mathbb{A}^1}$, at $t = 0$ it gives $s \circ p$.

Now suppose $a, b\colon \mathcal{F} \to \mathcal{G}$ are $\mathbb{A}^1$-homotopic via H. Since $p \circ i_0 = p \circ i_1 = \mathrm{id}$ and $\mathrm{L}_{\mathbb{A}^1}(p)$ is an isomorphism, applying $\mathrm{L}_{\mathbb{A}^1}$ gives $\mathrm{L}_{\mathbb{A}^1}(i_0) = \mathrm{L}_{\mathbb{A}^1}(i_1)$. Hence

$$\mathrm{L}_{\mathbb{A}^1}(a) = \mathrm{L}_{\mathbb{A}^1}(H) \circ \mathrm{L}_{\mathbb{A}^1}(i_0) = \mathrm{L}_{\mathbb{A}^1}(H) \circ \mathrm{L}_{\mathbb{A}^1}(i_1) = \mathrm{L}_{\mathbb{A}^1}(b).$$

The result follows from the definition of homotopy equivalence. □

1.4.1. Explicit formula for $\mathrm{L}_{\mathbb{A}^1}$

The $\mathbb{A}^1$-localization functor admits an explicit formula via an algebraic singular complex.

Definition 1.4.3 (Algebraic simplex)**.** The standard n-simplex over S is

$$\Delta^n_S \;:=\; \mathrm{Spec}\big(\mathcal{O}_S[T_0, \dots, T_n]/(T_0 + \cdots + T_n - 1)\big) \;\subset\; \mathbb{A}^{n+1}_S.$$

Face and degeneracy maps are defined as in topology, yielding a cosimplicial scheme $\Delta^\bullet_S\colon \mathbf{\Delta} \to \mathsf{Sch}_S$. Note $\Delta^n_S \simeq \mathbb{A}^n_S$ as S-schemes (but not as cosimplicial objects).

Theorem 1.4.4 (Explicit formula for $\mathbb{A}^1$-localization)**.** *For $\mathcal{F} \in \mathsf{PShv}(\mathsf{Sm}_S)$ and $X \in \mathsf{Sm}_S$,*

$$(\mathrm{L}_{\mathbb{A}^1}\mathcal{F})(X) \;\simeq\; \mathrm{colim}_{[n]\in\mathbf{\Delta}^{\mathrm{op}}} \mathcal{F}(X \times \Delta^n_S).$$

Proof. Write $\mathcal{F}'(X) := \mathrm{colim}_{[n]\in\mathbf{\Delta}^{\mathrm{op}}} \mathcal{F}(X \times \Delta^n_S)$.

$\mathbb{A}^1$**-invariance.** The projection $p\colon X \times \mathbb{A}^1 \to X$ and zero section s induce maps on the cosimplicial objects $X\times\Delta^\bullet_S$ and $X\times\mathbb{A}^1\times\Delta^\bullet_S$. We have $s^*\circ p^* = \mathrm{id}$, and the multiplication $\mu\colon \mathbb{A}^1\times\mathbb{A}^1 \to \mathbb{A}^1$ defines an $\mathbb{A}^1$-homotopy $p^*s^* \simeq \mathrm{id}$. Since $\Delta^1_S \simeq \mathbb{A}^1_S$, this $\mathbb{A}^1$-homotopy translates into a simplicial homotopy on $\mathcal{F}(X\times\Delta^\bullet_S)$ between p^*s^* and id, which becomes an equivalence after $\mathrm{colim}_{\mathbf{\Delta}^{\mathrm{op}}}$. Hence $p^*\colon \mathcal{F}'(X) \xrightarrow{\sim} \mathcal{F}'(X \times \mathbb{A}^1)$.

Universal property. Let $\mathcal{G}$ be $\mathbb{A}^1$-invariant. Since $\Delta^n_S \simeq \mathbb{A}^n_S$, $\mathbb{A}^1$-invariance forces $\mathcal{G}(X \times \Delta^\bullet_S)$ to have all simplicial structure maps acting as equivalences; hence $\mathrm{colim}_{\mathbf{\Delta}^{\mathrm{op}}} \mathcal{G}(X \times \Delta^\bullet_S) \simeq \mathcal{G}(X)$. Therefore

$$\mathrm{Hom}(\mathcal{F}', \mathcal{G}) \simeq \lim_{\mathbf{\Delta}} \mathrm{Hom}(\mathcal{F}(- \times \Delta^n_S), \mathcal{G}) \simeq \mathrm{Hom}\big(\mathcal{F}, \lim_{\mathbf{\Delta}} \mathcal{G}(- \times \Delta^n_S)\big) \simeq \mathrm{Hom}(\mathcal{F}, \mathcal{G}).$$ □

***Remark* 1.4.5** (Geometric meaning)**.** Theorem 1.4.4 is the algebro-geometric analogue of the topological singular complex $\mathrm{Sing}_\bullet$. The simplicial anima $\mathcal{F}(X \times \Delta^\bullet_S)$ encodes algebraic homotopy information: 0-simplices are sections, 1-simplices parametrize $\mathbb{A}^1$-paths, 2-simplices parametrize higher homotopies. Taking the colimit over $\mathbf{\Delta}^{\mathrm{op}}$ quotients out all $\mathbb{A}^1$-deformations.

Caveat: $\mathrm{L}_{\mathbb{A}^1}$ does not generally preserve Nisnevich sheaves, so the motivic localization $\mathrm{L}_{\mathrm{mot}}$ is *not* simply $\mathrm{L}_{\mathbb{A}^1} \circ \mathrm{L}_{\mathrm{Nis}}$; see Remark 1.5.3.

1.4.2. Interval objects

We have encountered three a priori distinct notions: $\mathbb{A}^1$-localization, localization at $\mathbb{A}^1$-homotopy equivalences, and the $\mathrm{L}_{\mathbb{A}^1}$-construction via $\Delta^\bullet$. Do they coincide? Following Hoyois [Hoy], the answer is yes, and this is a formal consequence of $\mathbb{A}^1$ being an **interval object**.

Definition 1.4.6 (Bipointed object). Let $\mathcal{C}$ be a category with finite products. A **bipointed object** is an object $I \in \mathcal{C}$ with two global sections $0, 1 \colon * \rightrightarrows I$.

Definition 1.4.7 (Three homotopy categories). Let $\mathcal{C}$ have finite products and I be a bipointed object.

(i) **I-localization** (nullification): formally invert all projections $X \times I \to X$:

$$\mathcal{C}/I \;:=\; \mathcal{C}\left[\{X \times I \to X\}^{-1}\right].$$

(ii) **Localization at I-homotopy equivalences**:

$$\mathcal{C}\left[\mathrm{hoeq}_I^{-1}\right].$$

(iii) **I-homotopy category**, via simplicial enrichment:

$$\mathrm{Ho}(\mathcal{C}, I), \qquad \mathrm{Hom}_{\mathrm{Ho}(\mathcal{C},I)}(X, Y) = \mathrm{colim}_{\mathbf{\Delta}^{\mathrm{op}}} \mathrm{Hom}_{\mathcal{C}}(X \times I^n, Y).$$

In general, there are canonical functors

$$\mathcal{C}\left[\mathrm{hoeq}_I^{-1}\right] \longrightarrow \mathrm{Ho}(\mathcal{C}, I) \longrightarrow \mathcal{C}/I,$$

neither of which is generally an equivalence.

Definition 1.4.8 (Interval object). A bipointed object $I \in \mathcal{C}$ is an **interval object** if

(i) $X \times I$ exists for all $X \in \mathcal{C}$;

(ii) I is I-contractible: $I \to *$ is an I-homotopy equivalence.

Example 1.4.9.

- The unit interval $[0, 1] \in \mathsf{Top}$ with multiplication $\mu(x, y) = xy$.
- The affine line $\mathbb{A}^1 \in \mathsf{Sm}_S$ with ring multiplication.
- The real line $\mathbb{R}$ and complex plane $\mathbb{C}$ in Top, both with multiplication.

Theorem 1.4.10 (Morel–Voevodsky [MV99]). *For any category $\mathcal{C}$ and interval object $I \in \mathcal{C}$,*

$$\mathcal{C}\left[\mathrm{hoeq}_I^{-1}\right] \xrightarrow{\sim} \mathrm{Ho}(\mathcal{C}, I) \xrightarrow{\sim} \mathcal{C}/I.$$

Proof sketch. **Left equivalence.** I-contractibility provides a zigzag of I-homotopies connecting id_I to the constant map $0 \colon I \to * \to I$. Taking products with X shows that $(\mathrm{id}_X, 0) \colon X \to X \times I$ is an I-homotopy inverse to pr_1, so every projection $X \times I \to X$ is an I-homotopy equivalence.

Right equivalence. The I-homotopy from id_I to 0 induces a simplicial homotopy between the corresponding endomorphisms of $\mathrm{Hom}(I \times I^\bullet, Y)$. After $\mathrm{colim}_{\mathbf{\Delta}^{\mathrm{op}}}$ this becomes an isomorphism, identifying mapping anima in $\mathrm{Ho}(\mathcal{C}, I)$ with those in $\mathcal{C}/I$. □

Corollary 1.4.11. *The affine line $\mathbb{A}^1$ with sections $0, 1$ and ring multiplication is an interval object in Sm_S and in $\mathcal{P}_\Sigma(\mathsf{Sm}_S)$. Hence the three a priori distinct notions in §1.4 coincide on $\mathcal{P}_\Sigma(\mathsf{Sm}_S)$:*

(i) $\mathbb{A}^1$-localization (inverting $\mathcal{F} \times \mathbb{A}^1 \to \mathcal{F}$);

(ii) localization at $\mathbb{A}^1$-homotopy equivalences;

(iii) the singular complex $\mathrm{L}_{\mathbb{A}^1}$.

***Remark* 1.4.12.** The multiplication $\mu(x, y) = xy$ on $\mathbb{A}^1$ provides the $\mathbb{A}^1$-homotopy from $\mathrm{id}_{\mathbb{A}^1}$ (at $y = 1$) to the constant map 0 (at $y = 0$). This is why 1, not 0, plays the role of the basepoint: the unit of the multiplication is the structure that makes $\mathbb{A}^1$ an interval object.

1.5. Motivic Spaces

We now assemble the construction. Let S be qcqs. Since Sm_S is extensive, $\mathcal{P}_\Sigma(\mathsf{Sm}_S) \simeq \mathsf{Shv}_\sqcup(\mathsf{Sm}_S)$, and finite coproduct decompositions are Nisnevich covers, so $\mathsf{Shv}_{\mathrm{Nis}}(\mathsf{Sm}_S) \subset \mathcal{P}_\Sigma(\mathsf{Sm}_S)$.

Definition 1.5.1 (Motivic spaces). The category of **motivic spaces** over S is

$$\mathsf{H}(S) \;:=\; \mathsf{Shv}_{\mathrm{Nis}}(\mathsf{Sm}_S) \cap \mathsf{PShv}_{\mathbb{A}^1}(\mathsf{Sm}_S) \;\subset\; \mathsf{PShv}(\mathsf{Sm}_S),$$

consisting of presheaves of anima satisfying both Nisnevich descent and $\mathbb{A}^1$-invariance. The corresponding localization functor is

$$\mathrm{L}_{\mathrm{mot}} \colon \mathsf{PShv}(\mathsf{Sm}_S) \longrightarrow \mathsf{H}(S).$$

A morphism in $\mathsf{PShv}(\mathsf{Sm}_S)$ is a **motivic equivalence** if $\mathrm{L}_{\mathrm{mot}}$ sends it to an equivalence.

Fact 1.5.2. *(i) $\mathrm{L}_{\mathrm{Nis}}$ is left exact. Both $\mathrm{L}_{\mathbb{A}^1}$ and $\mathrm{L}_{\mathrm{mot}}$ preserve finite products [Hoy14, Proposition C.6].*

(ii) Motivic equivalences form the smallest strongly saturated class in $\mathsf{PShv}(\mathsf{Sm}_S)$ containing both Nisnevich-local and $\mathbb{A}^1$-local equivalences.

***Remark* 1.5.3** (Non-commutativity of localizations). In general, neither $\mathrm{L}_{\mathrm{mot}} = \mathrm{L}_{\mathbb{A}^1} \circ \mathrm{L}_{\mathrm{Nis}}$ nor $\mathrm{L}_{\mathrm{mot}} = \mathrm{L}_{\mathrm{Nis}} \circ \mathrm{L}_{\mathbb{A}^1}$ holds: $\mathrm{L}_{\mathbb{A}^1}$ can destroy Nisnevich descent, and $\mathrm{L}_{\mathrm{Nis}}$ can destroy $\mathbb{A}^1$-invariance. But $\mathsf{H}(S)$ is defined as an intersection, independent of any order of localization. Concretely, the motivic localization is computed by ping-ponging:

$$\mathrm{L}_{\mathrm{mot}} \;\simeq\; (\mathrm{L}_{\mathrm{Nis}} \circ \mathrm{L}_{\mathbb{A}^1})^{\circ \mathbb{N}},$$

i.e., transfinite alternation until stabilization.

1.5.1. $\mathbb{A}^1$-rigidity

Definition 1.5.4 ($\mathbb{A}^1$-rigidity). An S-scheme X is **$\mathbb{A}^1$-rigid** if $\mathrm{L}_{\mathbb{A}^1}(X) \simeq X$ in $\mathcal{P}_\Sigma(\mathsf{Sm}_S)$, i.e., $X(U \times_S \mathbb{A}^1) \simeq X(U)$ for all $U \in \mathsf{Sm}_S$.

Proposition 1.5.5. *The following are $\mathbb{A}^1$-rigid:*

(i) the multiplicative group $\mathbb{G}_m$ over any reduced scheme of finite Krull dimension;

(ii) smooth projective curves of genus ≥ 1 over a field;

(iii) abelian varieties.

Proof. For $\mathbb{G}_m$ over a reduced base S: for $U \in \mathsf{Sm}_S$ (hence reduced), $\mathbb{G}_m(U \times \mathbb{A}^1) = \mathcal{O}(U \times \mathbb{A}^1)^\times = \mathcal{O}(U)[t]^\times = \mathcal{O}(U)^\times = \mathbb{G}_m(U)$, where the third equality uses that the units of a polynomial ring over a reduced ring are just the constant units. For smooth projective curves of genus ≥ 1 and abelian varieties, this follows from the fact that there are no non-constant morphisms from $\mathbb{A}^1$ to a proper variety of non-zero dimension with no rational curves. □

1.5.2. Motivic spaces and K-theory

Connective K-theory $\mathrm{k} \colon \mathsf{Sm}_S^{\mathrm{op}} \to \mathsf{An}$ satisfies Nisnevich descent (Example 1.2.12) and is $\mathbb{A}^1$-invariant when the base is regular, hence defines an object of $\mathsf{H}(S)$.

Theorem 1.5.6 (Representability of K-theory). *For S regular Noetherian of finite Krull dimension,*

$$\mathrm{L}_{\mathrm{mot}}(\mathbb{Z} \times \mathrm{BGL}_\infty) \;\simeq\; \mathrm{k}.$$

In particular, for affine $U \in \mathsf{Sm}_S$, $[U, \mathbb{Z} \times \mathrm{BGL}_\infty]_{\mathsf{H}(S)} \simeq \mathrm{K}_0(U)$.

***Remark* 1.5.7** (Affine representability). Although Vect_r is not $\mathbb{A}^1$-invariant on all smooth schemes (Remark 1.1.1), motivic localization still yields the correct values on smooth affine schemes: Asok–Hoyois–Wendt [AHW17] proved that if X is a presheaf on Sm_S such that $\pi_0(X)$ is $\mathbb{A}^1$-invariant on affine schemes and X satisfies affine Nisnevich excision, then $\pi_0(X)(U) \xrightarrow{\sim} [U, X]_{\mathsf{H}(S)}$ for all smooth affine U. Taking $X = \mathrm{BGL}_r$ gives $\pi_0 \mathsf{Vect}_r(U) \simeq [U, \mathrm{BGL}_r]_{\mathsf{H}(S)}$ for affine U.

***Remark* 1.5.8** (The role of regularity). Nisnevich descent of K-theory holds without regularity (Example 1.2.12); $\mathbb{A}^1$-invariance does not. For regular Noetherian X, Quillen proved $\mathrm{K}_n(X) \xrightarrow{\sim} \mathrm{K}_n(X \times \mathbb{A}^1)$ for all $n \geq 0$. For singular schemes this fails: with $R = k[\varepsilon]/(\varepsilon^2)$,

$$\mathrm{K}_1(R[t]) \neq \mathrm{K}_1(R),$$

since $(1 + \varepsilon t) \in R[t]^\times$ contributes a class in $\mathrm{K}_1(R[t])$ not coming from $\mathrm{K}_1(R)$.

To restore $\mathbb{A}^1$-invariance on non-regular schemes, Weibel introduced **homotopy K-theory**

$$\mathrm{KH}(X) \;:=\; \mathrm{colim}_{[n]\in\mathbf{\Delta}^{\mathrm{op}}}\, \mathrm{K}\big(X \times \Delta^n\big),$$

which is precisely $\mathrm{L}_{\mathbb{A}^1}$ applied to K.

1.6. Pointed Motivic Spaces

1.6.1. Basic constructions

Definition 1.6.1 (Pointed motivic spaces). The category $\mathsf{H}_*(S)$ of **pointed motivic spaces** is the category of pointed objects in $\mathsf{H}(S)$.

The functor $(-)_+ \colon \mathsf{H}(S) \to \mathsf{H}_*(S)$ adds a disjoint basepoint, $X_+ = X \sqcup *$, and is left adjoint to the forgetful functor.

The **smash product** on $\mathsf{H}_*(S)$ is the pushout

$$X \wedge Y \;:=\; \mathrm{cofib}(X \vee Y \to X \times Y),$$

making $\mathsf{H}_*(S)$ symmetric monoidal with unit S_+.

We adopt the basepoint conventions:

- $\mathbb{A}^1$ is pointed at 1;
- $\mathbb{G}_m$ is pointed at 1;
- $\mathbb{P}^1$ is pointed at $\infty = [1 : 0]$.

Definition 1.6.2 (Suspension and loops). For $X \in \mathsf{H}_*(S)$,

- the **suspension** $\Sigma X := S^1 \wedge X$, where $S^1 = [1]/\partial[1]$ is the simplicial circle (as a constant sheaf);
- the **loop space** $\Omega X := \mathrm{fib}(* \to X)$.

These form an adjunction $\Sigma \dashv \Omega$.

1.6.2. Bigraded spheres

In $\mathsf{H}_*(S)$ there are two essentially different "circles".

Definition 1.6.3 (Motivic spheres).

- The **simplicial circle** is $S^{1,0} := S^1$ (constant sheaf).
- The **Tate circle** is $S^{1,1} := \mathbb{G}_m$ (pointed at 1).
- The **motivic (p, q)-sphere**, for $p \geq q \geq 0$, is

$$S^{p,q} \;:=\; (S^{1,0})^{\wedge(p-q)} \wedge (S^{1,1})^{\wedge q}.$$

The key computation:

Proposition 1.6.4 ($\mathbb{P}^1$ as a motivic sphere). *In* $\mathsf{H}_*(S)$, $\mathbb{P}^1 \simeq S^{2,1} = S^1 \wedge \mathbb{G}_m$.

Proof. Consider the standard Nisnevich square for $\mathbb{P}^1$:

$$\begin{tikzcd} \mathbb{G}_m \arrow[r] \arrow[d] & \mathbb{A}^1 \arrow[d] \\ \mathbb{A}^1 \arrow[r] & \mathbb{P}^1. \end{tikzcd}$$

By Brown–Gersten in the pointed setting, this is a pushout square in $\mathsf{H}_*(S)$. Since $\mathbb{A}^1 \simeq *$ in $\mathsf{H}(S)$, both copies of $\mathbb{A}^1$ collapse to the basepoint, and the pushout becomes

$$\mathbb{P}^1 \;\simeq\; * \sqcup_{\mathbb{G}_m} * \;\simeq\; \Sigma\mathbb{G}_m \;=\; S^1 \wedge \mathbb{G}_m \;=\; S^{2,1}. \qquad \square$$

***Remark* 1.6.5.** This makes precise the observation that $\mathbb{G}_m$ realizes a kind of "algebraic circle": under Betti realization $\mathbb{G}_m(\mathbb{C}) = \mathbb{C}^\times$ is homotopy equivalent to the topological S^1. The motivic sphere $\mathbb{G}_m$ plays the role of $S^{1,1}$, while the simplicial circle S^1 is $S^{1,0}$. Their smash product $\mathbb{P}^1 \simeq S^{2,1}$ is the natural "algebraic 2-sphere".

This bigraded structure is what makes motivic homotopy theory richer than classical homotopy theory: the weight direction q encodes Tate twists specific to algebraic geometry.

Proposition 1.6.6 (Punctured affine spaces). *For $n \geq 1$, $\mathbb{A}^n \setminus \{0\} \simeq S^{2n-1,n}$ in $\mathsf{H}_*(S)$ (with any choice of basepoint, e.g. $(1, 0, \ldots, 0)$).*

Proof. By induction on n. The case $n = 1$ is $\mathbb{A}^1 \setminus \{0\} = \mathbb{G}_m \simeq S^{1,1}$. For $n \geq 2$, the Nisnevich square

$$\begin{tikzcd} (\mathbb{A}^{n-1} \setminus \{0\}) \times \mathbb{G}_m \arrow[r] \arrow[d] & \mathbb{A}^{n-1} \times \mathbb{G}_m \arrow[d] \\ (\mathbb{A}^{n-1} \setminus \{0\}) \times \mathbb{A}^1 \arrow[r] & \mathbb{A}^n \setminus \{0\} \end{tikzcd}$$

is a pushout in $\mathsf{H}_*(S)$. Since $\mathbb{A}^1 \simeq *$, the right two terms collapse, yielding

$$\mathbb{A}^n \setminus \{0\} \;\simeq\; \Sigma\big((\mathbb{A}^{n-1} \setminus \{0\}) \wedge \mathbb{G}_m\big) \;=\; S^{1,0} \wedge S^{2n-3,n-1} \wedge S^{1,1} \;=\; S^{2n-1,n}. \qquad \square$$

Corollary 1.6.7 (Thom spaces). *In $\mathsf{H}_*(S)$,*

$$\mathbb{A}^n/(\mathbb{A}^n \setminus \{0\}) \;\simeq\; S^{2n,n} \quad \textit{and} \quad \mathbb{P}^n/\mathbb{P}^{n-1} \;\simeq\; S^{2n,n}.$$

Proof. The cofiber sequence $\mathbb{A}^n \setminus \{0\} \to \mathbb{A}^n \to \mathbb{A}^n/(\mathbb{A}^n \setminus \{0\})$ together with $\mathbb{A}^n \simeq *$ gives

$$\mathbb{A}^n/(\mathbb{A}^n \setminus \{0\}) \;\simeq\; \Sigma(\mathbb{A}^n \setminus \{0\}) \;\simeq\; \Sigma S^{2n-1,n} \;=\; S^{2n,n}.$$

For the second, use the standard open cover of $\mathbb{P}^n$ and induction. $\square$

1.7. Norms on Motivic Spaces

We close the chapter by sketching the construction of **norm functors** due to Bachmann–Hoyois [BH21]. These provide additional multiplicative structure on motivic homotopy theory and play a central role in the theory of motivic spectra developed in Chapter 2.

1.7.1. Motivation and statement

For a finite locally free morphism $p\colon T \to S$ in classical algebraic geometry, there is the **Weil restriction** $\mathrm{R}_p\colon \mathsf{Sm}_T \to \mathsf{Sm}_S$, right adjoint to pullback $p^*\colon \mathsf{Sm}_S \to \mathsf{Sm}_T$, characterized by

$$\mathrm{Hom}_S(Y, \mathrm{R}_p X) \;\simeq\; \mathrm{Hom}_T(Y \times_S T, X).$$

In the pointed setting, Weil restriction extends to a symmetric monoidal functor $p_\otimes\colon \mathsf{Sm}_{T,+} \to \mathsf{Sm}_{S,+}$ (with respect to the smash product). The core question of [BH21] is: can $p_\otimes$ be extended to motivic homotopy theory?

Theorem 1.7.1 (Bachmann–Hoyois [BH21, Theorems 3.3, 4.1]). *Let $p\colon T \to S$ be a finite locally free, universally open morphism. Then there exists a symmetric monoidal functor*

$$p_{\otimes}\colon \mathsf{H}_*(T) \longrightarrow \mathsf{H}_*(S)$$

satisfying:

(i) *$p_{\otimes}$ preserves sifted colimits;*

(ii) *$p_{\otimes}(X_+) \simeq (\mathrm{R}_p X)_+$ for $X \in \mathsf{Sm}_T$;*

(iii) *if $T = S^{\sqcup n}$ and p is the fold map, then $p_{\otimes}$ is the n-fold smash product;*

(iv) *if p is moreover finite étale, $p_{\otimes}$ extends to a symmetric monoidal functor $p_{\otimes}\colon \mathsf{SH}(T) \to \mathsf{SH}(S)$.*

***Remark* 1.7.2** (Universal property of $\mathcal{P}_\Sigma$). The proof of Theorem 1.7.1 factors the construction through the universal sequence

$$\mathsf{Sm}_{T,+} \xrightarrow{(1)} \mathcal{P}_\Sigma(\mathsf{Sm}_T)_* \xrightarrow{(2)} \mathsf{H}_*(T) \xrightarrow{(3)} \mathsf{SH}(T).$$

Each step is a universal construction in $\mathrm{CAlg}(\mathsf{Cat}^{\mathrm{sift}})$ (presentably symmetric monoidal categories with sifted colimits). Consequently, the symmetric monoidal $p_{\otimes}\colon \mathsf{Sm}_{T,+} \to \mathsf{Sm}_{S,+}$ from Weil restriction has at most one sifted-colimit-preserving symmetric monoidal extension at each stage:

(i) the extension to $\mathcal{P}_\Sigma$ is unconditional;

(ii) the extension to H_* holds iff $p_{\otimes}$ preserves Nisnevich and $\mathbb{A}^1$ equivalences (which is the case for finite locally free universally open p);

(iii) the extension to SH holds iff $p_{\otimes}(\mathbb{P}^1_{T,+})$ is $\otimes$-invertible in $\mathsf{SH}(S)$ (which is the case for finite étale p).

This is a paradigmatic example of how the universal-property approach to motivic homotopy theory pays off: a hard "can we extend?" question reduces to checking elementary preservation properties.

***Remark* 1.7.3.** The functor $p_{\otimes}$ preserves neither arbitrary colimits nor arbitrary limits, so it cannot fit into a six-functor formalism. It is part of a richer "parametrized" structure on motivic spectra; see Chapter 2 and [BH21, §5–6].

This completes our treatment of unstable motivic homotopy theory. In Chapter 2, we turn to the stable setting: the category $\mathsf{SH}(S)$ obtained by inverting $\mathbb{T}_S = \mathbb{P}^1_S/\{\infty\}$ under the smash product.

Chapter 2.

Stable Motivic Homotopy Theory

2.1. Grothendieck's Vision and the Universal Property of SH

Algebraic geometry abounds with cohomology theories: ℓ-adic cohomology, crystalline cohomology, de Rham cohomology, Betti cohomology, and so on. Grothendieck envisioned that behind all these theories lies a **universal coefficient system** governing all cohomology theories on schemes.

More precisely, he conjectured the existence of an abelian category $\mathsf{Mot}(S)$ and a functor

$$\mathrm{Mot}\colon \mathsf{Sch}_S^{\mathrm{op}} \to \mathsf{Mot}(S)$$

such that for any "cohomology functor" $h^*\colon \mathsf{Sch}_S^{\mathrm{op}} \to \mathcal{A}$, there exists a unique **realization functor** $\bar{h}^*\colon \mathsf{Mot}(S) \to \mathcal{A}$ making the following diagram commute:

$$\begin{tikzcd} \mathsf{Sch}_S^{\mathrm{op}} \arrow[r, "h^*"] \arrow[dr, "\mathrm{Mot}"'] & \mathcal{A} \\ & \mathsf{Mot}(S) \arrow[u, dashed, "\bar{h}^*"'] \end{tikzcd}$$

In other words, $\mathsf{Mot}(S)$ would be the universal target for all cohomology theories. At that time, stable categories had not yet entered the mathematical landscape, so Grothendieck's vision was formulated in the language of abelian categories.

Today, stable categories have largely supplanted classical homological algebra, and "studying cohomology on schemes" is increasingly understood as "studying **six-functor formalisms** on schemes." Grothendieck's quest for universal coefficients thus naturally reformulates as:

Does there exist a universal six-functor formalism?

The answer depends on what axioms we impose. Different choices of axioms yield different "universal" objects. The motivic six-functor formalism SH arises from a specific list of axioms, formulated precisely by Drew–Gallauer [DG22].

Theorem 2.1.1 (Drew–Gallauer [DG22]; cf. [Sch26, Theorem 11.1]). *Let k be a noetherian base ring with $\mathrm{Spec}(k)$ of finite Krull dimension, and let C be the category of separated finite-type k-schemes, equipped with I = open immersions and P = proper morphisms. Consider the category of presentable six-functor formalisms*

$$\mathsf{D}\colon \mathsf{Corr}(C) \longrightarrow \mathsf{Pr}^L_{\mathrm{st}}$$

in which open immersions are cohomologically étale, proper maps are cohomologically proper, and which satisfy:

(i) ***Cohomological smoothness:*** *every smooth morphism is cohomologically smooth in* D*;*

(ii) ***Excision:*** *for any closed immersion $i\colon Z \hookrightarrow X$ with open complement $j\colon U \hookrightarrow X$, the sequence*

$$j_!\mathbf{1} \longrightarrow \mathbf{1} \longrightarrow i_*\mathbf{1}$$

is a fiber sequence in $\mathsf{D}(X)$*;*

(iii) $\mathbb{A}^1$***-homology trivial:*** *for $\pi\colon \mathbb{A}^1 \to \mathrm{Spec}(k)$, the unit map $\mathbf{1} \to \pi_!\pi^!\mathbf{1}$ is an equivalence.*

This category has an initial object, and that initial object is the motivic six-functor formalism SH *of Morel–Voevodsky.*

The proof of [DG22] is constructive, and recovers the construction of SH that we will give in §2.2: enforcing the existence of left adjoints $f_\sharp$ to smooth pullbacks gives the stable presheaf category $\mathsf{PShv}(\mathsf{Sm}_{/X}; \mathsf{Sp})$; excision implies Nisnevich descent and forces passage to $\mathsf{Shv}_{\mathrm{Nis}}(\mathsf{Sm}_{/X}; \mathsf{Sp})$; triviality of $\mathbb{A}^1$-homology forces $\mathbb{A}^1$-localization, yielding $\mathsf{Shv}_{\mathrm{Nis},\mathbb{A}^1}(\mathsf{Sm}_{/X}; \mathsf{Sp})$; finally, $\otimes$-inverting the Tate object yields SH. We refer the reader to [Sch26, §11] for a clean exposition of the proof sketch.

***Remark* 2.1.2** (Realizations of motives)**.** The power of Theorem 2.1.1 lies in its corollary: any other six-functor formalism on C satisfying axioms (1)–(3) automatically receives a unique symmetric monoidal functor from SH, called its **realization**. Concrete examples include:

- ℓ-adic étale cohomology $\mathsf{D}_{\text{ét}}(-, \mathbb{Z}_\ell)$, when ℓ is invertible on the base;
- Betti realization $\mathsf{D}(-(\mathbb{C}), \mathbb{Z})$, for $\mathbb{C}$-schemes;
- de Rham realization in characteristic 0.

Each yields a symmetric monoidal functor from SH and identifies the corresponding cohomology theory as a representable spectrum (the ℓ-adic, Betti, or de Rham realization spectrum). This is the modern conceptual explanation for why "every" reasonable cohomology theory of varieties admits motives in the first place.

However, SH is *not* the ultimate answer to Grothendieck's question. The axioms above—especially $\mathbb{A}^1$-invariance—exclude many important cohomology theories. In particular:

- Crystalline cohomology in characteristic p is not $\mathbb{A}^1$-invariant.
- Prismatic cohomology does not factor through SH.
- More generally, p-adic cohomology theories in characteristic p require fundamentally different frameworks.

Thus $\mathsf{SH}(S)$ should be understood as an *approximation* to Grothendieck's vision—the universal coefficient system for $\mathbb{A}^1$-invariant cohomology theories—rather than the definitive solution. The search for a truly universal framework encompassing all cohomology theories, including p-adic ones, remains an active and central problem in arithmetic geometry.

2.1.1. Six-functor formalisms via span categories

Before turning to the construction of $\mathsf{SH}(S)$, we briefly record the modern framework in which six-functor formalisms are now formulated, since it will clarify what we are ultimately aiming at.

Classically, a six-functor formalism on a category C of geometric objects is a packet of data $(f^*, f_*, f_!, f^!, \otimes, \underline{\mathrm{Hom}})$ subject to an intricate web of adjunctions, base change isomorphisms, and projection formula. Even writing down this data in a fully coherent categorical sense is a non-trivial task.

A far more economical description was initiated by Liu–Zheng [LZ24] and Gaitsgory–Rozenblyum [**?**], and brought to its present form by Mann [Man22] and Scholze [**?**]. Given a wide subcategory $E \subset C$ closed under composition and base change, one forms the **span category** $\mathsf{Span}(C, E)$: its objects are those of C, and its morphisms $X \to Y$ are spans

$$X \longleftarrow Z \xrightarrow{e} Y, \qquad e \in E,$$

composed by pullback. When C has finite products, $\mathsf{Span}(C, E)$ inherits a symmetric monoidal structure. In this language, a **three-functor formalism** on (C, E) is simply a lax symmetric monoidal functor

$$\mathsf{D} \colon \mathsf{Span}(C, E) \longrightarrow \mathsf{Cat},$$

which encodes f^* (backward legs), $f_!$ (forward legs), and $\otimes$ (monoidal structure) in a single coherent package. The remaining three operations f_*, $f^!$, $\underline{\mathrm{Hom}}$ are then obtained as right adjoints, and are therefore *properties* of D rather than additional structure.

The challenge is that constructing a lax monoidal functor out of $\mathsf{Span}(C, E)$ directly is hard. In practice, one starts from a much simpler piece of data, namely a functor

$$\mathsf{D}_0 \colon C^{\mathrm{op}} \longrightarrow \mathrm{CAlg}(\mathsf{Cat})$$

encoding only f^* and $\otimes$, and asks when it extends to $\mathsf{Span}(\mathcal{C}, E)$. The key input is a **suitable decomposition** of E in the sense of Mann [Man22, Def. A.5.9]: a pair of wide subcategories $I, P \subset E$ (thought of as "open/étale-like" and "proper-like" morphisms respectively) such that every $e \in E$ factors as $e = p \circ i$ with $i \in I$ and $p \in P$, subject to mild cancellation and truncation axioms. The extension theorem then reads: if $i^* = \mathsf{D}_0(i)$ admits a left adjoint $i_\sharp$ and $p^* = \mathsf{D}_0(p)$ admits a right adjoint p_* satisfying the expected base change and projection formulae—crucially including the mixed Beck–Chevalley equivalence $i_\sharp p'_* \simeq p_* i'_\sharp$—then D_0 admits an essentially unique extension to a three-functor formalism D, in which

$$f_! = p_* \, i_\sharp \qquad \text{for any factorization } f = p \circ i.$$

Recently, Cnossen–Lenz–Linskens [CLL25] gave a conceptually streamlined account of this extension theorem: they construct an 2-categorical enhancement $\mathsf{Span}_2(\mathcal{C}, E)_{P,I}$ of $\mathsf{Span}(\mathcal{C}, E)$—whose 2-morphisms are spans of spans with legs in P and I—and show that the inclusion $\mathcal{C}^{\mathrm{op}} \hookrightarrow \mathsf{Span}_2(\mathcal{C}, E)_{P,I}$ is the universal (I, P)-biadjointable functor. Mann's result becomes a formal corollary, and as a byproduct one obtains 2-categorical coherence data for D (in particular the adjoints $i_\sharp \dashv i^* \dashv i_*$ and $p^* \dashv p_*$ become part of the structure rather than auxiliary choices).

This is the framework in which we will place SH. Taking $\mathcal{C} = \mathsf{Sch}^{\mathrm{qcqs}}$, E = morphisms locally of finite presentation, I = étale morphisms, and P = proper morphisms, the six-functor formalism on SH will be obtained by verifying that the pullback functor $f^* \colon \mathsf{SH}(X) \to \mathsf{SH}(Y)$ and the smash product on SH satisfy precisely the hypotheses above. The deep input—smooth base change and proper base change for SH—is classical work of Morel–Voevodsky [MV99] and Ayoub [Ayo07] / Cisinski–Déglise [CD19]; the extension to the full six-functor package is then formal.

2.2. Construction of Motivic Spectra

This section gives a careful construction of the stable motivic category $\mathsf{SH}(S)$ as a universal $\otimes$-inversion, together with the accompanying technology—symmetric sequences, free algebras, Bousfield localization—needed both for this construction and for the construction of specific motivic spectra in subsequent sections. The treatment follows Robalo [Rob15], Hoyois [Hoy17], and Bachmann–Hoyois [BH21]; we have also drawn on the cleaner exposition of Annala–Iwasa [AI25] where appropriate.

Before entering the construction, let us briefly recall the setting established in Chapter 1. For a quasi-compact quasi-separated scheme S, we built $\mathsf{H}(S)$ as the full subcategory

$$\mathsf{H}(S) \;=\; \mathsf{Shv}_{\mathrm{Nis}}(\mathsf{Sm}_S) \cap \mathsf{PShv}_{\mathbb{A}^1}(\mathsf{Sm}_S) \;\subset\; \mathsf{PShv}(\mathsf{Sm}_S)$$

of presheaves (valued in anima) satisfying both Nisnevich descent and $\mathbb{A}^1$-invariance. The corresponding localization functor was denoted

$$\mathrm{L}_{\mathrm{mot}} \colon \mathsf{PShv}(\mathsf{Sm}_S) \longrightarrow \mathsf{H}(S),$$

and we showed (Remark 1.5.3) that it is computed as the transfinite alternation $\mathrm{L}_{\mathrm{mot}} \simeq (\mathrm{L}_{\mathrm{Nis}} \circ \mathrm{L}_{\mathbb{A}^1})^{\circ \mathbb{N}}$. The pointed variant $\mathsf{H}_*(S)$ was obtained by passing to pointed objects, and we defined the Tate object

$$\mathbb{T}_S \;:=\; \mathrm{cofib}\big(\infty \colon S \hookrightarrow \mathbb{P}^1_S\big) \;\in\; \mathsf{H}_*(S),$$

which—by Proposition 1.6.4—is equivalent to $S^1 \wedge \mathbb{G}_m = S^{2,1}$.

The goal of this section is to invert $\mathbb{T}_S$ under the smash product, producing a presentably symmetric monoidal category $\mathsf{SH}(S)$ that extends the sequence

$$\mathsf{PShv}(\mathsf{Sm}_S) \xrightarrow{\mathrm{L}_{\mathrm{mot}}} \mathsf{H}(S) \xrightarrow{(-)_+} \mathsf{H}_*(S) \longrightarrow \mathsf{SH}(S)$$

from Chapter 1 by one further universal construction.

2.2.1. The universal $\otimes$-inversion problem

Let $\mathcal{C} \in \mathrm{CAlg}(\mathsf{Pr}^L)$ be a presentably symmetric monoidal category and let $T \in \mathcal{C}$. We seek the initial presentably symmetric monoidal $\mathcal{C}$-algebra in which the image of T becomes $\otimes$-invertible: that is, the initial object of $\mathrm{CAlg}(\mathsf{Pr}^L)_{\mathcal{C}/}$ whose structure functor $\mathcal{C} \to \mathcal{D}$ sends T to an invertible object.

Proposition 2.2.1 (Robalo [Rob15, Proposition 2.9]). *There is a smashing localization*

$$(-)[T^{-1}]\colon \mathrm{Mod}_{\mathcal{C}}(\mathsf{Pr}^L) \longrightarrow \mathrm{Mod}_{\mathcal{C}}(\mathsf{Pr}^L)$$

whose essential image is spanned by those $\mathcal{D}$ on which T acts as an equivalence. The localization $\mathcal{C}[T^{-1}]$ acquires a unique presentably symmetric monoidal structure for which $\mathcal{C} \to \mathcal{C}[T^{-1}]$ is symmetric monoidal, and $\mathcal{D}[T^{-1}] \simeq \mathcal{C}[T^{-1}] \otimes_{\mathcal{C}} \mathcal{D}$ for every $\mathcal{D} \in \mathrm{Mod}_{\mathcal{C}}(\mathsf{Pr}^L)$.

This existence result is abstract; it gives no workable model of $\mathcal{C}[T^{-1}]$. To obtain one, we pass through symmetric sequences. The key point is that a sequential colimit model—the telescope

$$\mathsf{Tel}_T(\mathcal{D}) \coloneqq \mathrm{colim}\big(\mathcal{D} \xrightarrow{T\otimes -} \mathcal{D} \xrightarrow{T\otimes -} \cdots\big)$$

—fails to give the correct answer in $\mathrm{CAlg}(\mathsf{Pr}^L)$ in general: the telescope encodes only the cyclic $\mathbb{Z}$-action on $T^{\otimes n}$, whereas the universal property demands the full symmetric-group action. We must therefore use a construction that remembers the Σ_n-actions.

2.2.2. Symmetric sequences and free presentable algebras

Definition 2.2.2 (Symmetric sequences). Let $\mathbb{F}$ denote the groupoid of finite sets and bijections. For $\mathcal{C} \in \mathrm{CAlg}(\mathsf{Pr}^L)$, the category of **symmetric sequences** in $\mathcal{C}$ is

$$\mathsf{SSeq}(\mathcal{C}) \coloneqq \mathsf{Fun}(\mathbb{F}, \mathcal{C}),$$

equipped with the Day convolution symmetric monoidal structure induced by the disjoint-union monoidal structure on $\mathbb{F}$.

A symmetric sequence may be pictured as a list $X = (X_0, X_1, X_2, \ldots)$ with each X_n carrying a Σ_n-action. The Day convolution product is

$$(X \otimes Y)_n \;=\; \bigoplus_{p+q=n} \mathrm{Ind}_{\Sigma_p \times \Sigma_q}^{\Sigma_n}\big(X_p \otimes Y_q\big),$$

and the unit is $\mathbf{1}_{\mathsf{SSeq}(\mathcal{C})} = (\mathbf{1}_{\mathcal{C}}, \varnothing, \varnothing, \ldots)$.

Proposition 2.2.3 (Free $\mathcal{C}$-algebra on one object). *The symmetric sequence*

$$G_{\mathcal{C}} \;\coloneqq\; (\varnothing, \mathbf{1}_{\mathcal{C}}, \varnothing, \varnothing, \ldots) \;\in\; \mathsf{SSeq}(\mathcal{C}),$$

concentrated in degree 1, exhibits $\mathsf{SSeq}(\mathcal{C})$ as the free presentably symmetric monoidal $\mathcal{C}$-algebra on one generator. Concretely, for every $\mathcal{D} \in \mathrm{CAlg}(\mathsf{Pr}^L)_{\mathcal{C}/}$, evaluation at $G_{\mathcal{C}}$ induces an equivalence of anima

$$\mathrm{Hom}_{\mathrm{CAlg}(\mathsf{Pr}^L)_{\mathcal{C}/}}\big(\mathsf{SSeq}(\mathcal{C}), \mathcal{D}\big) \xrightarrow{\sim} \iota\mathcal{D}.$$

Proof sketch. This is a presentable categorical version of the classical fact that SSeq is the free symmetric monoidal cocompletion of the discrete category with one object. The key input is that $\mathbb{F}$ is the free symmetric monoidal anima on one generator, and that Day convolution with values in a presentable $\mathcal{C}$-algebra preserves colimits componentwise. See [Rob15, Lemma 2.3.1] or [BH21, Proposition C.1] for a detailed proof. □

The next piece of technology allows us to generate more intricate algebras by specifying a relation.

Definition 2.2.4 (Free symmetric algebra). For $X \in \mathcal{C}$, write

$$\mathrm{Sym}(X) \;\coloneqq\; \bigoplus_{n\geq 0} \big(X^{\otimes n}\big)_{h\Sigma_n} \;\in\; \mathrm{CAlg}(\mathcal{C})$$

for the free commutative algebra generated by X. It satisfies the universal property

$$\mathrm{Hom}_{\mathrm{CAlg}(\mathcal{C})}(\mathrm{Sym}(X), A) \;\simeq\; \mathrm{Hom}_{\mathcal{C}}(X, A) \qquad \text{for every } A \in \mathrm{CAlg}(\mathcal{C}).$$

Proposition 2.2.5 (Module categories as free pointed algebras). *For $X \in \mathcal{C}$, the module category $\mathrm{Mod}_{\mathrm{Sym}(X)}(\mathcal{C}) \in \mathrm{CAlg}(\mathsf{Pr}^L)_{\mathcal{C}/}$ is the initial presentably symmetric monoidal $\mathcal{C}$-algebra equipped with a morphism $X \to \mathbf{1}$ from (the image of) X to the structure unit. Concretely, for every $\mathcal{D} \in \mathrm{CAlg}(\mathsf{Pr}^L)_{\mathcal{C}/}$,*

$$\mathrm{Hom}_{\mathrm{CAlg}(\mathsf{Pr}^L)_{\mathcal{C}/}}\big(\mathrm{Mod}_{\mathrm{Sym}(X)}(\mathcal{C}), \mathcal{D}\big) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{D}}\big(X_{\mathcal{D}}, \mathbf{1}_{\mathcal{D}}\big),$$

where $X_{\mathcal{D}}$ denotes the image of X under the structure functor $\mathcal{C} \to \mathcal{D}$.

Proof sketch. By [Lur17, Theorem 4.8.5.16], maps of $\mathcal{C}$-algebras $\mathrm{Mod}_A(\mathcal{C}) \to \mathcal{D}$ correspond to maps of commutative algebras $A \to \mathbf{1}_{\mathcal{D}}$ in $\mathcal{C}$ (where $\mathbf{1}_{\mathcal{D}}$ is viewed as a $\mathcal{C}$-algebra via the structure functor). Applied to $A = \mathrm{Sym}(X)$, this becomes $\mathrm{Hom}_{\mathcal{C}}(X, \mathbf{1}_{\mathcal{D}})$, which under the structure functor identifies with $\mathrm{Hom}_{\mathcal{D}}(X_{\mathcal{D}}, \mathbf{1}_{\mathcal{D}})$. □

***Remark* 2.2.6.** The two universal properties (Propositions 2.2.3 and 2.2.5) are the only categorical inputs we will need. Everything that follows is a combination of these two and Bousfield localization.

2.2.3. Lax T-spectra

For $T \in \mathcal{C}$, consider the symmetric sequence

$$T_{\mathrm{sym}} \;:=\; (\varnothing, T, \varnothing, \varnothing, \ldots) \;\in\; \mathsf{SSeq}(\mathcal{C}),$$

concentrated in degree 1. Since $\mathsf{SSeq}(\mathcal{C})$ is itself a presentably symmetric monoidal $\mathcal{C}$-algebra, we may form its free commutative algebra $\mathrm{Sym}(T_{\mathrm{sym}}) \in \mathrm{CAlg}(\mathsf{SSeq}(\mathcal{C}))$, explicitly

$$\mathrm{Sym}(T_{\mathrm{sym}}) \;=\; \big(\mathbf{1}_{\mathcal{C}},\; T,\; (T^{\otimes 2})_{h\Sigma_2},\; (T^{\otimes 3})_{h\Sigma_3},\; \ldots\big),$$

with Σ_n-action on the n-th term given by the canonical permutation action.

Definition 2.2.7 (Lax T-spectra). The category of **lax T-spectra** in $\mathcal{C}$ is

$$\mathsf{Sp}_T^{\mathrm{lex}}(\mathcal{C}) \;:=\; \mathrm{Mod}_{\mathrm{Sym}(T_{\mathrm{sym}})}\big(\mathsf{SSeq}(\mathcal{C})\big) \;\in\; \mathrm{CAlg}(\mathsf{Pr}^L)_{\mathcal{C}/}.$$

Combining Propositions 2.2.3 and 2.2.5, we obtain:

Proposition 2.2.8 (Universal property of $\mathsf{Sp}_T^{\mathrm{lex}}$). *For every $\mathcal{D} \in \mathrm{CAlg}(\mathsf{Pr}^L)_{\mathcal{C}/}$ with structure functor $\varphi\colon \mathcal{C} \to \mathcal{D}$, there is a natural equivalence*

$$\mathrm{Hom}_{\mathrm{CAlg}(\mathsf{Pr}^L)_{\mathcal{C}/}}\big(\mathsf{Sp}_T^{\mathrm{lex}}(\mathcal{C}), \mathcal{D}\big) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{D}}\big(\varphi(T), \mathbf{1}_{\mathcal{D}}\big).$$

That is, $\mathsf{Sp}_T^{\mathrm{lex}}(\mathcal{C})$ is the initial presentably symmetric monoidal $\mathcal{C}$-algebra equipped with a morphism $T \to \mathbf{1}$.

Explicitly, an object of $\mathsf{Sp}_T^{\mathrm{lex}}(\mathcal{C})$ is a symmetric sequence $E = (E_0, E_1, E_2, \ldots)$ together with Σ_n-equivariant **bonding maps**

$$T \otimes E_n \longrightarrow E_{n+1}$$

(equivalently, by $\otimes$-adjunction, maps $E_n \to \mathrm{Hom}(T, E_{n+1})$) satisfying the associativity and $\Sigma_\bullet$-coherence expected of modules over $\mathrm{Sym}(T_{\mathrm{sym}})$.

2.2.4. T-spectra as a Bousfield localization

The universal property of $\mathsf{Sp}_T^{\mathrm{lex}}$ specifies a morphism $T \to \mathbf{1}$ (equivalently, a structure bonding map), but does not force it to be an equivalence. To turn T into an invertible object we must perform a further localization.

Definition 2.2.9 (T-spectra). The category of **T-spectra** in $\mathcal{C}$ is the $\otimes$-Bousfield localization

$$\mathrm{L}_{\mathsf{Sp}}\colon \mathsf{Sp}_T^{\mathrm{lex}}(\mathcal{C}) \longrightarrow \mathsf{Sp}_T(\mathcal{C})$$

at the class of bonding maps $T \otimes E \to \sigma_+ E$, where σ_+ is the shift operator $(\sigma_+ E)_n = E_{n-1}$. A lax T-spectrum E is a *T-spectrum* if and only if the adjoint bonding maps

$$E_n \xrightarrow{\sim} \mathrm{Hom}(T, E_{n+1})$$

are equivalences for all n.

Theorem 2.2.10 (Robalo [Rob15, Corollary 2.22]). *$\mathsf{Sp}_T(\mathcal{C})$ realizes the formal inversion of T:*

$$\mathsf{Sp}_T(\mathcal{C}) \simeq \mathcal{C}[T^{-1}] \qquad \text{in } \mathrm{CAlg}(\mathsf{Pr}^L)_{\mathcal{C}/}.$$

In particular, it is initial among presentably symmetric monoidal $\mathcal{C}$-algebras in which T becomes $\otimes$-invertible.

Theorem 2.2.11 (Telescope comparison [Rob15, Corollary 2.22]). *If T is **symmetric** in the sense that the cyclic permutation $\sigma\colon T^{\otimes 3} \to T^{\otimes 3}$, $a \otimes b \otimes c \mapsto b \otimes c \otimes a$, is homotopic to the identity in $\mathcal{C}$, then the canonical comparison functor*

$$\mathsf{Sp}_T(\mathcal{C}) \xrightarrow{\sim} \mathsf{Tel}_T(\mathcal{C}) = \mathrm{colim}\big(\mathcal{C} \xrightarrow{T\otimes -} \mathcal{C} \xrightarrow{T\otimes -} \cdots\big)$$

is an equivalence.

2.2.5. Two adjunctions and the explicit description

We now unwind what $\mathsf{Sp}_T(\mathcal{C})$ looks like as a concrete category. The structure functor factors as

$$\mathcal{C} \xrightarrow{\Sigma^\infty_T} \mathsf{Sp}_T(\mathcal{C}) \xrightarrow{\Omega^\infty_T} \mathcal{C},$$

with $\Sigma^\infty_T \dashv \Omega^\infty_T$, where $\Omega^\infty_T E = E_0$ and $\Sigma^\infty_T X$ is the "suspension spectrum" with $(\Sigma^\infty_T X)_n = T^{\otimes n} \otimes X$. More generally, for each $n \geq 0$ there is an adjunction

$$\Sigma^{\infty-n}_T \dashv \Omega^{\infty-n}_T, \qquad \Omega^{\infty-n}_T E = E_n.$$

Since T is invertible in $\mathsf{Sp}_T(\mathcal{C})$, these adjunctions extend to all $n \in \mathbb{Z}$, and we may define $E(n) := \Omega^{\infty-n}_T E$ for every integer n. The structure of a T-spectrum is then captured by the following:

Proposition 2.2.12 (Explicit description of T-spectra). *The data of an object $E \in \mathsf{Sp}_T(\mathcal{C})$ is equivalent to:*

- *a sequence of objects $\big(E(n)\big)_{n\in\mathbb{Z}}$ in $\mathcal{C}$;*
- *equivalences $c_n\colon E(n) \xrightarrow{\sim} \mathrm{Hom}(T, E(n+1))$ for every $n \in \mathbb{Z}$.*

A commutative algebra structure on E corresponds further to the data of compatible graded multiplications $E(m) \otimes E(n) \to E(m+n)$; equivalently, $(E, c) \in \mathrm{CAlg}(\mathsf{Sp}_T(\mathcal{C}))$ with $\Omega^{\infty-n}_T(E) = E(n)$ for all $n \in \mathbb{Z}$.

This description will be the principal working definition of a motivic spectrum in the applications below.

2.2.6. From graded algebras to motivic spectra

A common source of commutative algebras in $\mathsf{Sp}_T(\mathcal{C})$—including the motivic K-theory spectrum KGL and its relatives—is graded commutative algebras in $\mathcal{C}$ equipped with a distinguished *Bott element*. This subsection formalizes that passage.

Construction 2.2.13 (Graded algebras via $\mathsf{Fun}(\mathbb{Z}, \mathcal{C})$). Equip the set $\mathbb{Z}$ with its discrete groupoid structure, so that $\mathsf{Fun}(\mathbb{Z}, \mathcal{C}) \simeq \prod_{n\in\mathbb{Z}} \mathcal{C}$ acquires a symmetric monoidal structure via Day convolution for the addition operation on $\mathbb{Z}$. A **$\mathbb{Z}$-graded commutative algebra** in $\mathcal{C}$ is an element of $\mathrm{CAlg}(\mathsf{Fun}(\mathbb{Z}, \mathcal{C}))$; concretely, a sequence $E(*) = (E(n))_{n\in\mathbb{Z}}$ with graded multiplications $E(m) \otimes E(n) \to E(m+n)$ satisfying commutativity and associativity coherently.

The inclusion of groupoids $\mathbb{F} \hookrightarrow \mathbb{N} \hookrightarrow \mathbb{Z}$ (with $\mathbb{F}$ the groupoid of finite sets, $\mathbb{N}$ the discrete monoid of natural numbers) induces a lax symmetric monoidal functor

$$\mathsf{Fun}(\mathbb{Z}, \mathcal{C}) \xrightarrow{\text{forget}} \mathsf{SSeq}(\mathcal{C}),$$

restricting a graded object to its underlying symmetric sequence (forgetting the Σ_n-actions, which are trivial in the $\mathbb{Z}$-graded case).

Proposition 2.2.14 (Graded algebras with Bott element give lax spectra). *Let $E(*) \in \mathrm{CAlg}(\mathsf{Fun}(\mathbb{Z}, \mathcal{C}))$ be a $\mathbb{Z}$-graded commutative algebra in $\mathcal{C}$, and let $c\colon T \to E(1)$ be a morphism in $\mathcal{C}$ (a **Bott element**). Then the pair $(E(*), c)$ canonically promotes to a commutative algebra*

$$(E, c) \in \mathrm{CAlg}\big(\mathsf{Sp}_T^{\mathrm{lex}}(\mathcal{C})\big),$$

with underlying symmetric sequence the restriction of $E(*)$ and bonding maps $T \otimes E(n) \xrightarrow{c \otimes 1} E(1) \otimes E(n) \to E(n+1)$ given by multiplication.

Proof sketch. The Bott element c and the graded multiplication together determine a morphism $T_{\mathrm{sym}} \to E$ in $\mathsf{SSeq}(\mathcal{C})$ (using the forgetful functor from $\mathsf{Fun}(\mathbb{Z}, \mathcal{C})$); the universal property of $\mathrm{Sym}(T_{\mathrm{sym}})$ then makes E into a $\mathrm{Sym}(T_{\mathrm{sym}})$-module, i.e., a lax T-spectrum. □

The spectrum (E, c) constructed above lies in $\mathsf{Sp}_T^{\mathrm{lex}}$; to promote it to an honest T-spectrum, one needs the bonding maps to become equivalences. This is not automatic—it is a property of $(E(*), c)$—and one typically enforces it via the Bousfield localization of the next subsection.

2.2.7. Bott-periodic algebras

We now describe an alternative construction, due in this form to Bachmann–Hoyois [BH21], which directly produces objects of $\mathsf{Sp}_T(\mathcal{C})$ from pairs (E, β) where β is a Bott element. This construction is especially convenient when E is defined as a presheaf on schemes and one wishes to promote it to a motivic spectrum.

Definition 2.2.15 (Bott-periodic modules). Let $E \in \mathrm{CAlg}(\mathcal{C})$ and let $\beta\colon T \to E$ be a morphism in $\mathcal{C}$ (the **Bott element**). The adjoint of β is a morphism $\mathbf{1} \to \mathrm{Hom}(T, E)$, equivalently a morphism $E \to \mathrm{Hom}(T, E)$ of E-modules. The category of **β-periodic E-modules** is the $\otimes$-Bousfield localization

$$\mathcal{P}_\beta \mathrm{Mod}_E(\mathcal{C}) \subset \mathrm{Mod}_E(\mathcal{C})$$

spanned by those modules M for which the structure map $M \to \mathrm{Hom}(T, M)$ is an equivalence (equivalently, those on which $T \otimes -$ acts as an equivalence).

Proposition 2.2.16 (Promotion to T-spectra [BH21, §4]). *Let $E \in \mathrm{CAlg}(\mathcal{C})$ and $\beta\colon T \to E$ be as above. Then there is a natural equivalence*

$$\mathcal{P}_\beta \mathrm{Mod}_E(\mathcal{C}) \simeq \mathcal{P}_\beta \mathrm{Mod}_E\big(\mathsf{Sp}_T(\mathcal{C})\big),$$

where on the right E is viewed as a commutative algebra in $\mathsf{Sp}_T(\mathcal{C})$ via its image under Σ_T^∞. Consequently, E admits an essentially unique promotion to a commutative algebra

$$E \in \mathrm{CAlg}\big(\mathsf{Sp}_T(\mathcal{C})\big) \qquad \text{with } \Omega_T^{\infty - n} E = E \text{ for all } n \in \mathbb{Z}.$$

Informally: the Bott element β already encodes enough "symmetry" to force E to become T-periodic, and this periodicity is exactly what is needed for E to lift from $\mathcal{C}$ to $\mathsf{Sp}_T(\mathcal{C})$.

2.2.8. Specialization to motivic homotopy theory

We now apply the preceding machinery to the setting of motivic homotopy theory. The input data is the pointed motivic homotopy category $\mathsf{H}_*(S)$ from Chapter 1, equipped with the smash product, and the Tate object $\mathbb{T}_S \in \mathsf{H}_*(S)$. To use the machinery of this section, the only remaining verification is that $\mathbb{T}_S$ is symmetric in the sense of Theorem 2.2.11.

Proposition 2.2.17 ($\mathbb{T}_S$ is symmetric). *The cyclic permutation $\sigma\colon \mathbb{T}_S^{\wedge 3} \to \mathbb{T}_S^{\wedge 3}$ is $\mathbb{A}^1$-homotopic to the identity in $\mathsf{H}_*(S)$.*

Proof. By Proposition 1.6.4, $\mathbb{T}_S \simeq \mathbb{P}^1_S$. It suffices to show that a transposition on $\mathbb{P}^1 \wedge \mathbb{P}^1$ is $\mathbb{A}^1$-homotopic to $-\mathrm{id}$, for then a 3-cycle in Σ_3—being a composite of two transpositions—acts as $(-\mathrm{id})^2 = \mathrm{id}$.

Consider the action of $\mathrm{SL}_2(\mathbb{Z})$ on $\mathbb{P}^1 \wedge \mathbb{P}^1$ via its action on the two factors. We claim that the two matrices

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \qquad \text{and} \qquad \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

are $\mathbb{A}^1$-homotopic. Indeed, set

$$A \;:=\; \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \;=\; \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Then

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \cdot A \;=\; \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix},$$

and since $A \in \mathrm{SL}_2(\mathbb{Z})$ factors as a product of elementary matrices (each of which is $\mathbb{A}^1$-homotopic to the identity via $\begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}$, $t \in \mathbb{A}^1$), A itself is $\mathbb{A}^1$-homotopic to the identity. Thus the transposition τ on $\mathbb{P}^1 \wedge \mathbb{P}^1$ is $\mathbb{A}^1$-homotopic to $(-\mathrm{id}) \wedge \mathrm{id} \simeq -\mathrm{id}$, as claimed. □

Definition 2.2.18 (Stable motivic category). The **stable motivic category** over S is

$$\mathsf{SH}(S) \;:=\; \mathsf{Sp}_{\mathbb{T}_S}\big(\mathsf{H}_*(S)\big) \;=\; \mathsf{H}_*(S)\big[\mathbb{T}_S^{-1}\big] \;\in\; \mathrm{CAlg}(\mathsf{Pr}^L_{\mathrm{st}}).$$

By Theorem 2.2.11 and Proposition 2.2.17, it admits the explicit telescope model

$$\mathsf{SH}(S) \;\simeq\; \mathrm{colim}\Big(\mathsf{H}_*(S) \xrightarrow{-\wedge\mathbb{T}_S} \mathsf{H}_*(S) \xrightarrow{-\wedge\mathbb{T}_S} \cdots\Big).$$

Assigning $S \mapsto \mathsf{SH}(S)$ and $f : T \to S$ to the symmetric monoidal pullback $f^* : \mathsf{SH}(S) \to \mathsf{SH}(T)$ defines a functor

$$\mathsf{SH} \colon (\mathsf{Sch}^{\mathrm{qcqs}})^{\mathrm{op}} \longrightarrow \mathrm{CAlg}(\mathsf{Pr}^L_{\mathrm{st}}).$$

***Remark* 2.2.19** (Comparison with Chapter 1's construction chain). Putting together all the constructions from Chapter 1 and this section, we obtain the full sequence of symmetric monoidal localizations

$$\mathsf{PShv}(\mathsf{Sm}_S) \xrightarrow{\mathrm{L}_{\mathrm{mot}}} \mathsf{H}(S) \xrightarrow{(-)_+} \mathsf{H}_*(S) \xrightarrow{\Sigma^\infty_{\mathbb{T}}} \mathsf{SH}(S),$$

each realizing a universal construction in $\mathrm{CAlg}(\mathsf{Pr}^L)$:

- $\mathrm{L}_{\mathrm{mot}}$: free enforcement of Nisnevich descent and $\mathbb{A}^1$-invariance;
- $(-)_+$: free adjunction of a basepoint;
- $\Sigma^\infty_{\mathbb{T}}$: free inversion of $\mathbb{T}_S$ under the smash product.

Each stage is determined by a clean universal property, and the composition is the universal symmetric monoidal stable category attached to Sm_S that is Nisnevich-local, $\mathbb{A}^1$-invariant, and $\mathbb{T}$-stable. This decomposition is crucial for the construction of norm functors (see the end of Chapter 1): each of the three arrows is a universal construction in $\mathrm{CAlg}(\mathsf{Cat}^{\mathrm{sift}})$, so a functor defined at the level of $\mathsf{Sm}_{S,+}$ has at most one sifted-colimit-preserving symmetric monoidal extension at each stage.

***Remark* 2.2.20** (Stability). Since $\mathbb{T}_S \simeq S^1 \wedge \mathbb{G}_m$ in $\mathsf{H}_*(S)$, inverting $\mathbb{T}_S$ in particular inverts the simplicial circle S^1 (after one smash with $\mathbb{G}_m$); hence $\mathsf{SH}(S)$ is stable. The bigraded spheres $S^{p,q} = (S^1)^{\wedge(p-q)} \wedge \mathbb{G}_m^{\wedge q}$ from §1.6.2 refine the usual integer suspension, and both $\Sigma^{1,0}$ (simplicial) and $\Sigma^{1,1}$ (Tate) are invertible.

2.2.9. Example: the motivic K-theory spectrum KGL

We illustrate the Bott-periodic construction of §2.2.7 by building the motivic K-theory spectrum $\mathrm{KGL}_S \in \mathrm{CAlg}(\mathsf{SH}(S))$. This serves both as the prototype for the method and as a foundational object we will return to repeatedly.

Let $\mathrm{K} \colon \mathsf{Sm}_S^{\mathrm{op}} \to \mathsf{Sp}$ denote non-connective algebraic K-theory. Since K fails $\mathbb{A}^1$-invariance on singular schemes (Remark 1.5.8), we first replace it by homotopy K-theory

$$\mathrm{KH}(X) \;:=\; \mathrm{L}_{\mathbb{A}^1}\mathrm{K}(X) \;=\; \mathrm{colim}_{[n]\in\mathbf{\Delta}^{\mathrm{op}}}\, \mathrm{K}\big(X \times \Delta^n\big),$$

a Nisnevich sheaf of spectra on $\mathsf{Sch}^{\mathrm{qcqs}}$ satisfying $\mathbb{A}^1$-invariance. Restricted to Sm_S, KH defines an object of $\mathrm{CAlg}(\mathsf{H}(S); \mathsf{Sp})$ (the spectrum-valued analogue of $\mathsf{H}(S)$; we omit the notational refinement).

The Bott element arises geometrically: the fiber sequence $\mathcal{O}(-1) \to \mathcal{O} \to \mathcal{O}_\infty$ of vector bundles on $\mathbb{P}^1$ determines a class

$$\beta \in \mathrm{fib}\big(\mathrm{K}(\mathbb{P}^1) \to \mathrm{K}(\infty)\big),$$

the class of $[\mathcal{O}(-1)] - [\mathcal{O}]$. Using $\mathbb{T}_S \simeq \mathbb{P}^1_S/\{\infty\}$, this yields a morphism

$$\beta \colon \mathbb{T}_S \longrightarrow \mathrm{KH}$$

in $\mathsf{H}_*(S)$ (after passing to spectrum-valued presheaves).

Theorem 2.2.21 ([BH21, §16]). *The pair* (KH, β) *is Bott-periodic, and its promotion* $\mathrm{KGL}_S \in \mathrm{CAlg}(\mathsf{SH}(S))$ *via Proposition 2.2.16 is the* ***motivic K-theory spectrum****. It satisfies* $\Omega^{\infty-n}_{\mathbb{T}}\mathrm{KGL}_S \simeq \mathrm{KH}$ *for all* $n \in \mathbb{Z}$*, reflecting the Bott periodicity* $\Sigma_{\mathbb{T}}\mathrm{KGL}_S \simeq \mathrm{KGL}_S$*.*

Proposition 2.2.22 (Base change for KGL). *For any morphism of schemes* $f : Y \to X$*, the natural map*

$$f^*\mathrm{KGL}_X \xrightarrow{\sim} \mathrm{KGL}_Y$$

is an equivalence in $\mathsf{SH}(Y)$*.*

The proof relies on the following structural result on connective K-theory.

Proposition 2.2.23 (Bhatt–Lurie). *The connective K-theory functor*

$$\mathrm{K}_{\geq 0} \colon \mathrm{CRing}_R \longrightarrow \mathsf{Sp}$$

is left Kan extended from its restriction to the subcategory SmRing_R *of smooth R-algebras.*

Proof of Proposition 2.2.23. It suffices to show that the rank-n vector bundle stack $\mathsf{Vect}_n : \mathrm{CRing}_R \to \mathsf{An}$ is left Kan extended from SmRing_R, since K-theory of an animated commutative ring is built functorially from Vect_n. Recall that a presheaf is left Kan extended from a subcategory if and only if it can be written as a colimit of representables from that subcategory. Consider the chain of morphisms of stacks

$$\bigsqcup_k U_{n,k} \longrightarrow \bigsqcup_k \mathrm{Gr}_n(\mathbb{A}^k) \longrightarrow \mathsf{Vect}_n,$$

where $\mathrm{Gr}_n(\mathbb{A}^k)$ classifies rank-n subbundles of $\mathcal{O}^k$, and the $U_{n,k}$ are the standard smooth affine charts of $\mathrm{Gr}_n(\mathbb{A}^k)$ under the Plücker embedding. On affine schemes the composite is an effective epimorphism (every vector bundle embeds into a trivial bundle of sufficiently large rank), so Vect_n is the colimit of the Čech nerve of this composite. Each level of the Čech nerve is a disjoint union of smooth affine schemes—in other words, representable by objects of SmRing_R. Hence Vect_n, and therefore $\mathrm{K}_{\geq 0}$, is left Kan extended from SmRing_R. □

Proof of Proposition 2.2.22. The pullback functor $f^* \colon \mathsf{SH}(X) \to \mathsf{SH}(Y)$ is realized as a symmetric monoidal Bousfield localization of the corresponding functor on lax $\mathbb{T}$-spectra:

$$\mathsf{Sp}^{\mathrm{lex}}_T\big(\mathsf{PShv}(\mathsf{Sm}_X, \mathsf{An}_*)\big) \longrightarrow \mathsf{Sp}^{\mathrm{lex}}_T\big(\mathsf{PShv}(\mathsf{Sm}_Y, \mathsf{An}_*)\big),$$

so it suffices to verify the base-change equivalence already at the level of $\mathsf{Sp}^{\mathrm{lex}}_T(\mathsf{PShv}(\mathsf{Sm}_-, \mathsf{An}_*))$. Via the Bott-periodic construction (§2.2.7), this in turn reduces to showing $f^*\mathrm{KH}_X \simeq \mathrm{KH}_Y$ together with compatibility of Bott elements.

By Proposition 2.2.23, $\mathrm{K}_{\geq 0}$ is left Kan extended from smooth affine algebras; the same holds for non-connective K-theory K (by Bass delooping) and for KH (by $\mathbb{A}^1$-localization, which preserves left Kan extensions). Left Kan extension is automatically compatible with base change, completing the proof. □

2.3. The Six-Functor Formalism on SH

Having constructed SH as a functor $(\mathsf{Sch}^{\mathrm{qcqs}})^{\mathrm{op}} \to \mathrm{CAlg}(\mathsf{Pr}^L_{\mathrm{st}})$, we now extend it to a full six-functor formalism. We begin by recalling the relevant axioms.

2.3.1. Localization and base change

Definition 2.3.1 (Localization axiom). A functor $\mathsf{D}\colon (\mathsf{Sch}^{\mathrm{qcqs}})^{\mathrm{op}} \to \mathsf{Cat}_{\mathrm{st}}$ satisfies the **localization axiom** if for every $X \in \mathsf{Sch}^{\mathrm{qcqs}}$, every closed immersion $i\colon Z \hookrightarrow X$ with open complement $j\colon U \hookrightarrow X$, the diagram

$$\begin{tikzcd} \mathsf{D}(Z) \arrow[r, "i_*"] & \mathsf{D}(X) \arrow[l, bend right, "i^*"'] \arrow[l, bend left, "i^!"] \arrow[r, "j^*"] & \mathsf{D}(U) \arrow[l, bend right, "j_!"'] \arrow[l, bend left, "j_*"] \end{tikzcd}$$

is a recollement. That is, (i^*, j^*) is conservative, and both i_* and j_* are fully faithful.

Definition 2.3.2 (Base change properties). Let $\mathsf{D}\colon (\mathsf{Sch}^{\mathrm{qcqs}})^{\mathrm{op}} \to \mathsf{Cat}_{\mathrm{st}}$ be a functor.

- We say a morphism $g\colon C \to D$ satisfies **left base change** if for every pullback square

$$\begin{tikzcd} A \arrow[r, "k"] \arrow[d, "h"'] \arrow[dr, phantom, "\lrcorner", very near start] & B \arrow[d, "f"] \\ C \arrow[r, "g"'] & D \end{tikzcd}$$

  the functor $k^*\colon \mathsf{D}(B) \to \mathsf{D}(A)$ admits a left adjoint $k_\sharp$, and the canonical morphism $k_\sharp h^* \to f^* g_\sharp$ is an isomorphism.

- We say a morphism $f\colon B \to D$ satisfies **right base change** if for every pullback square as above, the functor $h^*\colon \mathsf{D}(C) \to \mathsf{D}(A)$ admits a right adjoint h_*, and the canonical morphism $g^* f_* \to h_* k^*$ is an isomorphism.

Proposition 2.3.3. *Let* $\mathsf{D}\colon (\mathsf{Sch}^{\mathrm{qcqs}})^{\mathrm{op}} \to \mathsf{Cat}_{\mathrm{st}}$ *satisfy the localization axiom.*

(i) If étale morphisms satisfy left base change, then D *satisfies Nisnevich descent.*

(ii) If finitely presented proper morphisms satisfy right base change, then D *satisfies abstract blowup descent.*

Proof. We treat the case of abstract blowup descent; the Nisnevich case is similar (and dual). Consider an abstract blowup square

$$\begin{tikzcd} E \arrow[r, "k"] \arrow[d, "q"'] \arrow[dr, phantom, "\lrcorner", very near start] & \tilde{X} \arrow[d, "p"] \\ Z \arrow[r, hook, "i"'] & X. \end{tikzcd}$$

By assumption, p^* and q^* admit right adjoints p_* and q_*. We must show that the functor

$$(p^*, i^*)\colon \mathsf{D}(X) \longrightarrow \mathsf{D}(\tilde{X}) \times_{\mathsf{D}(E)} \mathsf{D}(Z)$$

is an equivalence. Its right adjoint G sends a pair (A, B) with common image $C \in \mathsf{D}(E)$ to the pullback

$$G(A, B) \;=\; p_*A \times_{(iq)_*C} i_*B.$$

Conservativity. By the localization axiom, (i^*, j^*) is conservative, where $j\colon U = X \setminus Z \hookrightarrow X$. Since p restricts to an isomorphism over U, we have $j^* \simeq h^*p^*$ (with $h\colon U \hookrightarrow \tilde{X}$ the open complement of k), so (p^*, i^*) is also conservative.

Counit is an equivalence. It remains to show that for every (A, B), the counit map $(p^*, i^*) \circ G(A, B) \to (A, B)$ is an equivalence. We verify each component.

The i^*-component. Since i is a closed immersion, i_* is fully faithful by the localization axiom, so $i^*i_* \simeq \mathrm{id}$. Combined with right base change for p along i (giving $i^*p_*A \simeq q_*k^*A \simeq q_*C$), the fiber product becomes

$$i^*(p_*A \times_{(iq)_*C} i_*B) \;\simeq\; q_*C \times_{q_*C} B \;\simeq\; B.$$

The p^*-component. We must show $p^*(p_*A \times_{(iq)_*C} i_*B) \xrightarrow{\sim} A$. Since (k^*, h^*) is conservative on $\mathsf{D}(\tilde{X})$ by the localization axiom, we check after each.

Applying k^*: by $k^*p^* = q^*i^*$ and the previous case,

$$k^*p^*(p_*A \times_{(iq)_*C} i_*B) \;\simeq\; q^*B \;\simeq\; C \;\simeq\; k^*A.$$

Applying h^*: since i_*B and $(iq)_*C$ are supported on Z, applying $j^* = h^*p^*$ kills them, leaving j^*p_*A. We must show $j^*p_*A \simeq h^*A$. Consider the diagram of fiber sequences from the localization axiom:

$$\begin{tikzcd}
i_*i^!p_*A \arrow[r] \arrow[d] & p_*A \arrow[r] \arrow[d, equal] & j_*j^*p_*A \arrow[d] \\
p_*k_*k^!A \arrow[r] & p_*A \arrow[r] & p_*h_*h^*A.
\end{tikzcd}$$

The left vertical arrow is an equivalence by right base change for i along p. The middle arrow is the identity. Hence the right vertical arrow is also an equivalence; since $p_*h_* = j_*$ (because $ph = j$) and j_* is fully faithful, this gives $j^*p_*A \simeq h^*A$. □

2.3.2. The six-functor formalism for SH

Theorem 2.3.4 (Morel–Voevodsky [MV99]). *The stable motivic functor* $\mathsf{SH}\colon (\mathsf{Sch}^{\mathrm{qcqs}})^{\mathrm{op}} \to \mathsf{Pr}^L_{\mathrm{st}}$ *satisfies the localization axiom.*

Theorem 2.3.5 (Ayoub [Ayo07], Cisinski–Déglise [CD19]). *There is an essentially unique lax symmetric monoidal extension making the following diagram commute:*

$$\begin{tikzcd}
(\mathsf{Sch}^{\mathrm{qcqs}})^{\mathrm{op}} \arrow[r, "\mathsf{SH}"] \arrow[d, hook] & \mathrm{CAlg}(\mathsf{Pr}^L_{\mathrm{st}}) \\
\mathsf{Span}_2(\mathsf{Sch}^{\mathrm{qcqs}}, E)^{\otimes}_{P,I} \arrow[ur, dashed] &
\end{tikzcd}$$

with E = morphisms locally of finite presentation, I = étale morphisms, P = proper morphisms. The $(\infty, 2)$-category $\mathsf{Span}_2(\mathsf{Sch}^{\mathrm{qcqs}}, E)_{P,I}$ *has:*

- objects: *quasi-compact quasi-separated schemes;*
- 1-morphisms $X \to Z$: *spans*

$$\begin{tikzcd}
 & Y \arrow[dl, "f"'] \arrow[dr, "g"] & \\
X & & Z
\end{tikzcd}$$

  with f of finite presentation and g locally of finite presentation;

- 2-morphisms *between two such spans: diagrams*

$$\begin{tikzcd}
 & Y & \\
X \arrow[ur] & Y'' \arrow[u, "a"] \arrow[d, "b"] & Z \arrow[ul] \\
 & Y' \arrow[ul] \arrow[ur] &
\end{tikzcd}$$

  with a proper and b étale.

In this extension, the ***exceptional pushforward*** *along $f \in E$ is given by $f_! = p_*\, i_\sharp$ for any factorization $f = p \circ i$ with i étale and p proper.*

***Remark* 2.3.6** (Structure of the proof)**.** The proof is not a monolith; it decomposes into a formal step and a deep geometric step.

The *formal step* is the universal property of $\mathsf{Span}_2(-, E)^{\otimes}_{P,I}$ established by Cnossen–Lenz–Linskens [CLL25, Thm. 5.17]: given any lax symmetric monoidal functor $\mathsf{D}_0 \colon (\mathcal{C}^{\times})^{\mathrm{op}} \to \mathsf{Cat}^{\times}$ which is (I, P)-biadjointable and satisfies the projection formula, there is an essentially unique extension to $\mathsf{Span}_2(\mathcal{C}, E)^{\otimes}_{P,I}$. This step is purely 2-categorical and contains no algebro-geometric content.

The *geometric step* is to verify the input hypothesis, namely that SH is (I, P)-biadjointable. This breaks further into three ingredients:

- **Smooth/étale base change and the projection formula for $i_\sharp$:** these follow essentially from the construction of SH, since the smooth pushforward $i_\sharp$ on $\mathsf{PShv}(\mathsf{Sm}_-)$ is computed via the restriction along $\mathsf{Sm}_{/T} \to \mathsf{Sm}_{/S}$ for $i \colon T \to S$ étale, and this is compatible with base change at the level of presheaves.
- **Localization:** the recollement (i^*, j^*) for closed immersions i with open complement j. This is Theorem 2.3.4, due to Morel–Voevodsky [MV99].
- **Proper base change and the projection formula for p_*:** this is the deep input. Ayoub [Ayo07] proved proper base change for projective morphisms; combined with localization [MV99], this yields proper base change for all projective morphisms. Cisinski–Déglise [CD19] extended this to arbitrary proper morphisms using Chow's lemma and a Noetherian approximation argument.

Once all three ingredients are in hand, the full six-functor package on SH is obtained by invoking the universal property [CLL25]. The mixed Beck–Chevalley $i_\sharp p'_* \simeq p_* i'_\sharp$ is not an additional input: it follows from proper base change applied to a pullback square whose one edge is an étale morphism.

By Proposition 2.3.3 combined with Theorem 2.3.4 and the proper base change part of Remark 2.3.6, we obtain:

Corollary 2.3.7. *The functor* SH *is a cdh sheaf.*

Corollary 2.3.8 (Cisinski)**.** *The presheaf* KH *is a cdh sheaf on* $\mathsf{Sch}^{\mathrm{qcqs}}$.

Proof. By Corollary 2.3.7, SH is a cdh sheaf on $\mathsf{Sch}^{\mathrm{qcqs}}$. Combined with Proposition 2.2.22, this means that the assignment $X \mapsto \mathrm{KGL}_X \in \mathsf{SH}(X)$ is itself a cdh-local section of the cocartesian fibration $\int \mathsf{SH}$. Taking mapping anima out of the unit yields

$$\mathrm{KH}(X) \simeq \mathrm{Map}_{\mathsf{SH}(X)}(\mathbf{1}, \mathrm{KGL}_X),$$

which inherits cdh descent. □

Bibliography

[AHW17] Aravind Asok, Marc Hoyois, and Matthias Wendt. Affine representability results in $\mathbb{A}^1$-homotopy theory, I: vector bundles. *Duke Mathematical Journal*, 166(10):1923–1953, 2017.

[AI25] Toni Annala and Ryomei Iwasa. Motivic spectra and universality of k-theory, 2025.

[Ayo07] Joseph Ayoub. *Les six opérations de Grothendieck et le formalisme des cycles évanescents dans le monde motivique*. PhD thesis, Université Paris Diderot, 2007. Published as Astérisque **314–315** (2007).

[BEM25] Tom Bachmann, Elden Elmanto, and Matthew Morrow. $\mathbb{A}^1$-invariant motivic cohomology of schemes. 2025.

[BH21] Tom Bachmann and Marc Hoyois. Norms in motivic homotopy theory. *Astérisque*, 425:ix+207, 2021.

[CD19] Denis-Charles Cisinski and Frédéric Déglise. *Triangulated Categories of Mixed Motives*. Springer Monographs in Mathematics. Springer, 2019.

[CLL25] Bastiaan Cnossen, Tobias Lenz, and Sil Linskens. Universality of span 2-categories and the construction of 6-functor formalisms. 2025.

[DG22] Brad Drew and Martin Gallauer. The universal six-functor formalism. *Annals of K-Theory*, 7(4):599–649, December 2022.

[EM23] Elden Elmanto and Matthew Morrow. Motivic cohomology of equicharacteristic schemes. 2023. arXiv:2309.08463.

[Hoy] Marc Hoyois. The homotopy category of an interval object. Unpublished notes.

[Hoy14] Marc Hoyois. A quadratic refinement of the Grothendieck–Lefschetz–Verdier trace formula. *Algebraic & Geometric Topology*, 14(6):3603–3658, 2014.

[Hoy17] Marc Hoyois. The six operations in equivariant motivic homotopy theory. *Advances in Mathematics*, 305:197–279, January 2017.

[Lur17] Jacob Lurie. *Higher Algebra*. 2017.

[LZ24] Yifeng Liu and Weizhe Zheng. Enhanced six operations and base change theorem for higher artin stacks, 2024.

[Man22] Lucas Mann. A p-adic 6-functor formalism in rigid-analytic geometry, 2022.

[Mor12] Fabien Morel. *$\mathbb{A}^1$-Algebraic Topology over a Field*, volume 2052 of *Lecture Notes in Mathematics*. Springer, 2012.

[MV99] Fabien Morel and Vladimir Voevodsky. $\mathbb{A}^1$-homotopy theory of schemes. *Publications Mathématiques de l'IHÉS*, 90:45–143, 1999.

[Qui76] Daniel Quillen. Projective modules over polynomial rings. *Inventiones Mathematicae*, 36:167–171, 1976.

[Rob15] Marco Robalo. K-theory and the bridge from motives to noncommutative motives. *Advances in Mathematics*, 269:399–550, 2015.

[Sch26] Peter Scholze. Six-functor formalisms, 2026.

[Sus76] Andrei A. Suslin. Projective modules over polynomial rings are free. *Doklady Akademii Nauk SSSR*, 229(5):1063–1066, 1976.